

MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences
UNIST

Mathematical Analysis I & II

Course Textbook

Author

MUHO CHOI (*Undergraduate Student of CSE, UNIST*)

Email: sihyeon8240@unist.ac.kr

GitHub: <https://github.com/sihyeon8240>

Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/Math>

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Chapter 1

The Number Systems

Remark. Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \vee > \implies > \iff$$

Hence

$$\neg P \wedge Q \vee R \implies S \iff T$$

is understood as

$$(((\neg P) \wedge Q) \vee R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it unless parentheses indicate a smaller scope. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

whereas

$$(\forall x P(x)) \wedge (\forall x Q(x))$$

is different from

$$\forall x (P(x) \wedge Q(x))$$

1.1 The Natural Number Systems

The set $\{1, 2, 3, \dots\}$ of natural numbers is denoted by $\mathbb{N}$, and is defined by the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- If $S \subseteq \mathbb{N}$ satisfies
 - $1 \in S$
 - $\forall n \in S, s(n) \in S$

then $S = \mathbb{N}$

where $s(n)$ is called the **successor** of n . These axioms are called the **Peano axioms** (or *Peano Postulates*).

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** is not discussed, so we will just assume that $\mathbb{N}$ exists.

Definition 1.1.1. Let $P(n)$ be a statement about $n \in \mathbb{N}$, and let

$$S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\} \subseteq \mathbb{N}$$

Assume that $P(n)$ satisfies

- $P(1)$ is true
- $P(n)$ is true $\implies P(n + 1)$ is true

This implies that

- $1 \in S$
- $n \in S \implies n + 1 \in S$

Therefore,

$$\begin{aligned} S &= \mathbb{N} \\ \Downarrow \\ P(n) &\text{ is true for all } n \in \mathbb{N} \end{aligned}$$

This method of proof is called the **principle of mathematical induction**.

Definition 1.1.2. For all $n, m \in \mathbb{N}$, we define the **addition** operator $+$ on $\mathbb{N}$ by

- $n + 1 := s(n)$
- $n + s(m) := s(n + m)$

For all $n, m \in \mathbb{N}$, we define the **multiplication** operator $\cdot$ on $\mathbb{N}$ by

- $n \cdot 1 := n$
- $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, [Definition 1.1.2](#) requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operators on $\mathbb{N}$ are well-defined.

Also, we will not discuss the properties of the operators on $\mathbb{N}$, such as

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = n \cdot m + n \cdot k$

for all $n, m, k \in \mathbb{N}$. For these proofs rely on the recursive definitions.

Theorem 1.1.3. Let $S \subseteq \mathbb{N}$ be nonempty. Then S has a least element. This theorem is called the **well-ordering property** of $\mathbb{N}$.

Proof. Assume that S has no least element. Let

- $T := \mathbb{N} \setminus S$
- $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\}$

We will show that

$$\begin{array}{c}
 L = \mathbb{N} \\
 \Downarrow \\
 T = \mathbb{N} \\
 \Downarrow \\
 S = \emptyset \\
 \Downarrow \\
 \perp
 \end{array}$$

- **Base case:** If $1 \in S$, then

$$\begin{aligned} 1 &\in S \\ \Downarrow \\ 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} 1 &\notin S \\ \Downarrow \\ 1 &\in T \\ \Downarrow \\ 1 &\in L \end{aligned}$$

- **Inductive step:** Assume that $n \in L$ for some $n \in \mathbb{N}$. Then

$$\begin{aligned} n &\in L \\ \Downarrow \\ \{1, 2, \dots, n\} &\subseteq T \\ \Downarrow \\ \{1, 2, \dots, n\} \cap S &= \emptyset \end{aligned}$$

If $n + 1 \in S$, then

$$\begin{aligned} n + 1 &\in S \\ \Downarrow \\ n + 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} n + 1 &\notin S \\ \Downarrow \\ n + 1 &\in T \\ \Downarrow \\ \{1, 2, \dots, n + 1\} &\subseteq T \\ \Downarrow \\ n + 1 &\in L \end{aligned}$$

By the *principle of mathematical induction*, $L = \mathbb{N}$. □

1.2 Relations

Let S be a set. A **relation** $\sim$ on S is a subset of $S \times S$. Equivalently, a subset R of $S \times S$ is a relation on S .

- $R = \{(x, y) \in S \times S \mid x \sim y\}$

$$\bullet \ x \sim y \iff (x, y) \in R$$

Definition 1.2.1. Let S be a set. A relation $\sim$ on S is called an **equivalence relation** if it satisfies the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$
- **Symmetry:** $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity:** $\forall x, y, z \in S, (x \sim y \wedge y \sim z) \implies x \sim z$

Definition 1.2.2. Let S be a set. A collection $\{A_\alpha\}_{\alpha \in I}$ of sets is called a **partition** of S if

- $\forall \alpha \in I, (A_\alpha \neq \emptyset \wedge A_\alpha \subseteq S)$
- $\forall \alpha, \beta \in I, (\alpha \neq \beta \implies A_\alpha \cap A_\beta = \emptyset)$
- $\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.3. Let S be a set, and let $\sim$ be an equivalence relation on S . For each $x \in S$, the **equivalence class** of x is defined by

$$[x] := \{y \in S \mid y \sim x\}$$

We denote the set of all equivalence classes of S by

$$S/\sim := \{[x] \mid x \in S\}$$

Theorem 1.2.4. Let S be a set, and let $\sim$ be an equivalence relation on S . Then

$$\forall x, y \in S, [x] \cap [y] \neq \emptyset \implies [x] = [y]$$

Proof. Assume that $[x] \cap [y] \neq \emptyset$. Then

$$\begin{aligned} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \in [x] \wedge z \in [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{aligned}$$

Therefore, for any $w \in [x]$,

$$\begin{aligned} w &\sim x \wedge x \sim y \\ &\Downarrow \\ w &\sim y \\ &\Downarrow \\ w &\in [y] \end{aligned}$$

Therefore,

$$[x] \subseteq [y]$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.5. Let S be a set, and let $\{A_\alpha\}_{\alpha \in I}$ be a partition of S . Let

$$x \sim y \iff \exists \alpha \in I, (x \in A_\alpha \wedge y \in A_\alpha)$$

for all $x, y \in S$. Then

(A) $\sim$ is an equivalence relation on S

(B) $\{A_\alpha\}_{\alpha \in I} = S / \sim$

Proof. (A) Let $x, y, z \in S$.

• **Reflexivity:** For any x ,

$$\begin{aligned} x &\in S \\ &\Downarrow \\ x &\in \bigcup_{\alpha \in I} A_\alpha \\ &\Downarrow \\ \exists \alpha \in I, x &\in A_\alpha \\ &\Downarrow \\ x &\sim x \end{aligned}$$

• **Symmetry:** If $x \sim y$, then

$$\begin{aligned} \exists \alpha \in I, (x &\in A_\alpha \wedge y \in A_\alpha) \\ &\Downarrow \\ y &\in A_\alpha \wedge x \in A_\alpha \\ &\Downarrow \\ y &\sim x \end{aligned}$$

• **Transitivity:** If $x \sim y \wedge y \sim z$, then

$$\begin{aligned}
 & \exists \alpha \in I, (x \in A_\alpha \wedge y \in A_\alpha) \\
 & \exists \beta \in I, (y \in A_\beta \wedge z \in A_\beta) \\
 & \Downarrow \\
 & y \in A_\alpha \wedge y \in A_\beta \\
 & \Downarrow \\
 & A_\alpha \cap A_\beta \neq \emptyset \\
 & \Downarrow \\
 & \alpha = \beta \\
 & \Downarrow \\
 & x \in A_\alpha \wedge z \in A_\alpha \\
 & \Downarrow \\
 & x \sim z
 \end{aligned}$$

(B) Let $x \in S$. Then

$$\begin{aligned}
 & x \in S \\
 & \Downarrow \\
 & x \in \bigcup_{\alpha \in I} A_\alpha \\
 & \Downarrow \\
 & \exists \alpha \in I, x \in A_\alpha \\
 & \Downarrow \\
 & x \in [x] \\
 & \Downarrow \\
 & \exists \alpha \in I, A_\alpha = [x]
 \end{aligned}$$

Therefore,

$$\{A_\alpha\}_{\alpha \in I} \subseteq S / \sim$$

Let $[x] \in S / \sim$. Then

$$\begin{aligned}
 & x \in S \\
 & \Downarrow \\
 & x \in \bigcup_{\alpha \in I} A_\alpha \\
 & \Downarrow \\
 & \exists \alpha \in I, x \in A_\alpha \\
 & \Downarrow \\
 & A_\alpha \cap [x] \neq \emptyset \\
 & \Downarrow \\
 & A_\alpha = [x]
 \end{aligned}$$

Therefore,

$$S / \sim \subseteq \{A_\alpha\}_{\alpha \in I}$$

□

1.3 Ordered Sets

A set S is called an **ordered set** if S is equipped with a relation $<$ such that for any $x, y, z \in S$, the following properties hold:

$$(A) \quad x < y, y < z \implies x < z$$

$$(B) \quad \text{Only one of the relations } x < y, x = y, \text{ or } x > y \text{ holds}$$

the relation $<$ is called a **(strict) total order** on S .

On an ordered set, we define the following relations:

- $x > y \iff y < x$
- $x \geq y \iff (x > y \text{ or } x = y)$
- $x \leq y \iff y \geq x$

Example 1.3.1. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are ordered sets with the following relations:

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N} \text{ such that } n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+ \text{ such that } n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+ \text{ such that } p = q + r$

Definition 1.3.2. Let S be an ordered set and $E \subseteq S$.

- E is said to be **bounded above** if

$$\exists M \in S \text{ such that } \forall x \in E, x \leq M$$

Such an M is called an **upper bound** of E .

- E is said to be **bounded below** if

$$\exists m \in S \text{ such that } \forall x \in E, m \leq x$$

Such an m is called a **lower bound** of E .

Theorem 1.3.3. Let $E \subseteq \mathbb{Z}$ be nonempty.

(A) E is bounded below $\implies E$ has a least element.

(B) E is bounded above $\implies E$ has a greatest element.

Proof. (A) Let E be bounded below. Then,

$$\begin{aligned} &E \text{ is bounded below} \\ &\Downarrow \\ &\exists m \in \mathbb{Z} \text{ such that } \forall x \in E, m \leq x \end{aligned}$$

Let

$$F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} &E \neq \emptyset \\ &\Downarrow \\ &F \neq \emptyset \end{aligned}$$

By [Theorem 1.1.3](#), F has a least element, say n . Then

$$\begin{aligned} &\forall x \in E, n \leq x - m + 1 \\ &\Downarrow \\ &\forall x \in E, n + m - 1 \leq x \end{aligned}$$

Since $n = \min F$,

$$\begin{aligned} &n \in F \\ &\Downarrow \\ &\exists x_0 \in E \text{ such that } n = x_0 - m + 1 \\ &\Downarrow \\ &x_0 = n + m - 1 = \min E \end{aligned}$$

(B) Let E be bounded above. Then,

$$\begin{aligned} &E \text{ is bounded above} \\ &\Downarrow \\ &\exists M \in \mathbb{Z} \text{ such that } \forall x \in E, x \leq M \end{aligned}$$

Let

$$F := \{-x + M + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} &E \neq \emptyset \\ &\Downarrow \\ &F \neq \emptyset \end{aligned}$$

By [Theorem 1.1.3](#), F has a least element, say n . Then

$$\begin{aligned} &\forall x \in E, n \leq -x + M + 1 \\ &\Downarrow \\ &\forall x \in E, x \leq M - n + 1 \end{aligned}$$

Since $n = \min F$,

$$\begin{aligned} n &\in F \\ \Downarrow \\ \exists x_0 \in E \text{ such that } n &= -x_0 + M + 1 \\ \Downarrow \\ x_0 &= M - n + 1 = \max E \end{aligned}$$

□

Definition 1.3.4. Let S be an ordered set, and let $E \subseteq S$ be bounded above. An element $\alpha \in S$ is called the **supremum** (or *least upper bound*) of E if

- α is an upper bound of E
- $\forall \beta \in S, \beta < \alpha \implies \beta$ is not an upper bound of E

The supremum of E is denoted by

$$\alpha = \sup E$$

Similarly, let $E \subseteq S$ be bounded below. An element $\alpha \in S$ is called the **infimum** (or *greatest lower bound*) of E if

- α is a lower bound of E
- $\forall \beta \in S, \beta > \alpha \implies \beta$ is not a lower bound of E

The infimum of E is denoted by

$$\alpha = \inf E$$

Example 1.3.5. Let

$$S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$$

Let $p \in S$ be given, and let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$$

Then

$$\begin{aligned} q^2 - 2 &= \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \\ \Downarrow \\ q &\in S \end{aligned}$$

This implies that $p \in S$ is not an upper bound of S .

Let $p \in \mathbb{Q}^+ \setminus S$ be given. Then

$$p^2 \geq 2$$

$$\Downarrow \quad (\because \nexists p \in \mathbb{Q}^+, p^2 = 2)$$

$$p^2 > 2$$

$$\Downarrow \quad (\because \forall x \in S, x^2 < 2 < p^2 \implies x < p)$$

p is an upper bound of S

Let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$$

$$\Downarrow \quad (\because \forall x \in S, x^2 < 2 < q^2 \implies x < q)$$

q is an upper bound of S

This implies that p is not the least upper bound of S . To sum up, S has no least upper bound in $\mathbb{Q}^+$.

Definition 1.3.6. An ordered set S is said to have the **least upper bound property** if

$$\begin{aligned} \forall E \subseteq S, E \text{ is nonempty and bounded above} \\ \implies E \text{ has a least upper bound in } S. \end{aligned}$$

Theorem 1.3.7. Let S be an ordered set with the least upper bound property. Then

$$\begin{aligned} \forall E \subseteq S, E \text{ is nonempty and bounded below} \\ \implies E \text{ has a greatest lower bound in } S. \end{aligned}$$

Proof. Let E be a nonempty subset of S which is bounded below. Let L be defined as follows:

$$L := \{\lambda \mid \lambda \text{ is a lower bound of } E\}.$$

Then, L is nonempty and bounded above. By the least upper bound property, there exists $\alpha = \sup L \in S$. We will show that $\alpha = \inf E$.

Assume that $\alpha \notin L$. Then we have the following:

- α is not a lower bound of $E \implies$ There exists $\beta \in E$ such that $\beta < \alpha$.
- $\alpha = \sup L \implies \beta$ is not an upper bound of $L \implies$ There exists $\gamma \in L$ such that $\beta < \gamma \leq \alpha$.
- $\gamma \in L \implies \gamma$ is a lower bound of $E \implies \gamma \leq \beta$.

This is a contradiction. Therefore, we have $\alpha \in L$, i.e., α is a lower bound of E . Since α is an upper bound of L , the following holds:

For any $\delta > \alpha$, we have $\delta \notin L \implies \delta$ is not a lower bound of E .

Therefore, we can conclude that $\alpha = \inf E$. □

1.4 Fields and Ordered Fields

A set F is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$ for all $x, y \in F$.
- $x + y = y + x$ for all $x, y \in F$.
- $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- There exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For each $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$.

(B) Axioms for multiplication

- $x \cdot y \in F$ for all $x, y \in F$.
- $x \cdot y = y \cdot x$ for all $x, y \in F$.
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.
- There exists an element $1 \in F$ with $1 \neq 0$ such that $x \cdot 1 = x$ for all $x \in F$.
- For each $x \in F$ with $x \neq 0$, there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.

A set F is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If $x, y, z \in F$ and $x < y$, then $x + z < y + z$.
- If $x, y \in F$ and $x > 0$ and $y > 0$, then $x \cdot y > 0$.

Example 1.4.1. $(\mathbb{Q}, +, \cdot)$ is an ordered field, while $(\mathbb{Z}, +, \cdot)$ is not an ordered field since $2^{-1} \notin \mathbb{Z}$.

1.5 The Real Field

There exists an ordered field $(\mathbb{R}, +, \cdot)$ which has the *least upper bound property*. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield, i.e., $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication on $\mathbb{R}$ coincide with those on $\mathbb{Q}$.

We can construct $\mathbb{R}$ from $\mathbb{Q}$. The construction of $\mathbb{R}$ is rather complicated, so it will be discussed separately in [Appendix](#). In this section, we will just assume that $\mathbb{R}$ exists.

Theorem 1.5.1. *For any $x \in \mathbb{R}^+$ and $y \in \mathbb{R}$, there exists some $n \in \mathbb{N}$ such that $nx > y$. This theorem is called the **Archimedean property** of $\mathbb{R}$.*

Proof. Let $x \in \mathbb{R}^+$ and $y \in \mathbb{R}$ be given. Let E be defined as follows:

$$E := \{nx \mid nx \leq y, n \in \mathbb{N}\}.$$

If $E = \emptyset$, which implies that $x > y$, then $n = 1$ satisfies $nx > y$. Otherwise, E is nonempty and bounded above by y . By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}$. Then we have the following:

- $\alpha - x$ is not an upper bound of $E \implies$ There exists some $m \in \mathbb{N}$ such that $mx > \alpha - x$.
- $mx > \alpha - x \implies (m+1)x > \alpha \implies (m+1)x \notin E \implies (m+1)x > y$.

Therefore, we can take $n = m + 1$ to complete the proof. $\square$

Corollary 1.5.2. *For any $\varepsilon \in \mathbb{R}^+$ and $b \in (1, +\infty)$, there exists some $n \in \mathbb{N}$ such that $b^{-n} < \varepsilon$.*

Proof. For any $n \in \mathbb{N}$ and $x \in [-1, +\infty)$, we have

$$(1+x)^n \geq 1+nx.$$

Let $a \in \mathbb{R}^+$ and $\varepsilon \in \mathbb{R}^+$ be given. By [Theorem 1.5.1](#), there exists some $n \in \mathbb{N}$ such that $na > \frac{1}{\varepsilon}$. Then we have

$$(1+a)^n \geq 1+na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon.$$

Since $b > 1$, we can take $a = b - 1 > 0$ to complete the proof. $\square$

Theorem 1.5.3. *Any nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$ has a unique least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$ has a unique greatest element.*

Proof. Let E be a nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$. Then there exists a lower bound $\alpha \in \mathbb{R}$ of E . Then we have the following:

- If $\alpha \geq 0$, then $-1 \in \mathbb{Z}$ is a lower bound of E .
- If $\alpha < 0$, then by [Theorem 1.5.1](#), there exists some $n \in \mathbb{N}$ such that $n > -\alpha$, which implies that $-n \in \mathbb{Z}$ is a lower bound of E .

Therefore, E is bounded below in $\mathbb{Z}$. By [Theorem 1.3.3](#), E has a unique least element.

Let F be a nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$. Then there exists an upper bound $\beta \in \mathbb{R}$ of F . Then we have the following:

- By [Theorem 1.5.1](#), there exists some $n \in \mathbb{N}$ such that $n > \beta$, which implies that $n \in \mathbb{Z}$ is an upper bound of F .

Therefore, F is bounded above in $\mathbb{Z}$. By [Theorem 1.3.3](#), F has a unique greatest element. $\square$

Definition 1.5.4. For any $x \in \mathbb{R}$, the **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as follows:

- $\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$.
- $\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$.

[Theorem 1.5.3](#) guarantees the *existence* and *uniqueness* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.5.5. *For any $-\infty < x < y < +\infty$, there exists some $p \in \mathbb{Q}$ such that $x < p < y$. This theorem states that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.*

Proof. Let $\varepsilon := y - x > 0$. By [Theorem 1.5.1](#), there exists some $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. Let S be defined as follows:

$$S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} < x \right\}.$$

S is nonempty since $n\lfloor x \rfloor - 1 \in S$, and is bounded above by $nx \in \mathbb{R}$. By [Theorem 1.5.3](#), S has a greatest element, say m .

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take $p = \frac{m+1}{n} \in \mathbb{Q}$ to complete the proof. $\square$

Theorem 1.5.6. *For each $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists a unique $y \in \mathbb{R}^+$ such that $y^n = x$. We denote y by $\sqrt[n]{x}$ or $x^{\frac{1}{n}}$.*

Proof. Let E be defined as follows:

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\}.$$

E is nonempty since $2^{-1} \in E$ if $x \geq 1$, and $x^2 \in E$ otherwise. Also, E is bounded above by $\max\{x, 1\}$. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}^+$. We will show that $\alpha^n = x$.

For any $a, b \in \mathbb{R}^+$ with $a < b$, we have

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a)nb^{n-1}.$$

Assume that $\alpha^n < x$. By Theorem 1.5.5, there exists some $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}}\right\}.$$

Put $a = \alpha$ and $b = \alpha + h$. Then we have

$$\begin{aligned} (\alpha + h)^n - \alpha^n &< hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

Therefore, we have $\alpha + h \in E$, which is a contradiction.

Assume that $\alpha^n > x$. By Theorem 1.5.5, there exists some $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1, \alpha, \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}}\right\}.$$

Put $a = \alpha - h$ and $b = \alpha$. Then we have

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha - h)^n &> \alpha^n - hn(\alpha + 1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

Therefore, we have $\alpha - h$ is an upper bound of E , which is a contradiction.

Therefore, we have $\alpha^n = x$, which guarantees the *existence* of y . For any $\alpha, \beta \in \mathbb{R}^+$, we have $\alpha \neq \beta \implies \alpha^n \neq \beta^n$, which guarantees the *uniqueness* of y . $\square$

Definition 1.5.7. Let $y \in [0, 1)$ and $b \in \mathbb{N}$ be given with $b \geq 2$. A sequence $(a_n)_{n=1}^\infty$ is called the **greedy base b expansion** of y if for each $n \in \mathbb{N}$, we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

where $a_k \in S_b := \{0, 1, 2, \dots, b-1\}$ for all $k \in \mathbb{N}$. Especially, if $b = 10$, the sequence is called the **greedy decimal expansion** of y .

Theorem 1.5.8. For any $y \in [0, 1)$, the greedy base b expansion of y exists and is unique.

Proof. Let $(t_n)_{n=1}^\infty$, $(T_n)_{n=1}^\infty$ and $(a_n)_{n=1}^\infty$ be defined as follows:

- $t_1 := by$, $T_1 := \{t \in S_b \mid t \leq t_1\}$, $a_1 := \max T_1$.
 - $t_2 := b^2y - ba_1$, $T_2 := \{t \in S_b \mid t \leq t_2\}$, $a_2 := \max T_2$.
 - $\vdots$
 - $t_n := b^n y - \sum_{k=1}^{n-1} a_k b^{n-k}$, $T_n := \{t \in S_b \mid t \leq t_n\}$, $a_n := \max T_n$
- for all $n \in \mathbb{N}$ with $n \geq 2$,

Assume that $0 \leq t_n < b$ for some $n \in \mathbb{N}$. Then, since $0 \in T_n$ and T_n is bounded above, T_n has a greatest element by [Theorem 1.3.3](#), which implies that $(a_n)_{n=1}^\infty$ is well-defined. Let $P(n)$ be the statement that

$$P(n) : 0 \leq t_n < b \text{ and } a_n \text{ is well-defined.}$$

- **Base case:** $P(1)$ is true since $0 \leq t_1 = by < b$.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\ &= b \left(b^n y - \sum_{k=1}^n a_k b^{n-k} \right) \\ &= b \left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\ &= b(t_n - a_n). \end{aligned}$$

Since $a_n = \max T_n$ and $t_n < b$, we have

$$a_n \leq t_n < a_n + 1 \implies 0 \leq t_n - a_n < 1 \implies 0 \leq t_{n+1} = b(t_n - a_n) < b.$$

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. By the definition of a_n , we have

$$\begin{aligned} a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \implies 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\ \implies 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

for each $n \in \mathbb{N}$. Therefore, the *existence* of the *greedy base b expansion* of y is guaranteed.

Next, assume that there exist two sequences $(a_n)_{n=1}^\infty$ and $(c_n)_{n=1}^\infty$ such that for each $n \in \mathbb{N}$, we have

- $0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$
- $0 \leq y - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n}$

where $a_k \in S_b$ and $c_k \in S_b$ for all $k = 1, 2, \dots, n$. We will show that for each $n \in \mathbb{N}$, we have $a_n = c_n$. Let $Q(n)$ be the statement that

$$Q(n) : a_k = c_k \text{ for all } k = 1, 2, \dots, n.$$

• **Base case:** We have

- $0 \leq y - \frac{a_1}{b} < \frac{1}{b} \implies \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b}.$
- $0 \leq y - \frac{c_1}{b} < \frac{1}{b} \implies \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b}.$

By combining the two inequalities, we have

- $\frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} \implies a_1 < c_1 + 1 \implies a_1 \leq c_1.$
- $\frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} \implies c_1 < a_1 + 1 \implies c_1 \leq a_1.$

Therefore, we have $a_1 = c_1$, which shows that $Q(1)$ is true.

• **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. We have

- $0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}}.$

$$- 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}}.$$

By combining the two inequalities, we have

$$\begin{aligned} - \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}}. \\ - \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}}. \end{aligned}$$

By the inductive hypothesis, we have $a_k = c_k$ for all $k = 1, 2, \dots, n$. Therefore, we have

$$\begin{aligned} - \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies a_{n+1} - c_{n+1} < 1 \implies a_{n+1} \leq c_{n+1}. \\ - \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies c_{n+1} - a_{n+1} < 1 \implies c_{n+1} \leq a_{n+1}. \end{aligned}$$

Therefore, we have $a_{n+1} = c_{n+1}$, which shows that $Q(n+1)$ is true whenever $Q(n)$ is true.

By the *principle of mathematical induction*, $Q(n)$ is true for all $n \in \mathbb{N}$. Therefore, the *uniqueness* of the *greedy base b expansion* of y is guaranteed. $\square$

Theorem 1.5.9. Let $b \in \mathbb{N}$ be given with $b \geq 2$. Let $(a_n)_{n=1}^\infty$ be the greedy base b expansion of $a \in [0, 1)$, and let $(c_n)_{n=1}^\infty$ be the greedy base b expansion of $c \in [0, 1)$. If $a_n = c_n$ for all $n \in \mathbb{N}$, then we have $a = c$.

Proof. Assume that $a \neq c$. Without loss of generality, assume that $a < c$. Then, by [Corollary 1.5.2](#), there exists some $n \in \mathbb{N}$ such that

$$\frac{1}{b^n} < c - a.$$

By the definition of the *greedy base b expansion*, we have

$$\begin{aligned} \bullet \quad 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} &\implies \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n}. \\ \bullet \quad 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} &\implies c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}. \end{aligned}$$

By the assumption that $a_n = c_n$ for all $n \in \mathbb{N}$, we have

$$\sum_{k=1}^n \frac{a_k}{b^k} = \sum_{k=1}^n \frac{c_k}{b^k}.$$

Therefore, we have a contradiction:

$$\sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}.$$

Therefore, we have $a = c$, which completes the proof. $\square$

Theorem 1.5.10. *Let $b \in \mathbb{N}$ be given with $b \geq 2$, and let $(a_n)_{n=1}^\infty$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists no $y \in [0, 1)$ such that*

$$(a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y,$$

then there exists some $N \in \mathbb{N}$ such that

$$a_n = b - 1 \quad \text{for all } n \geq N.$$

Proof. We will prove the contrapositive of the statement: For all $N \in \mathbb{N}$, if there exists some $n \in \mathbb{N}$ such that

$$n \geq N \text{ and } a_n \neq b - 1,$$

then there exists $y \in [0, 1)$ such that $(a_n)_{n=1}^\infty$ is the greedy base b expansion of y .

Assume that the condition of the above claim holds. Let $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Since $a_k \in S_b$ for all $k \in \mathbb{N}$, we have

$$0 \leq x_n \leq \sum_{k=1}^n \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1 \quad \text{for all } n \in \mathbb{N}.$$

Then, we have that $X := \{x_n \mid n \in \mathbb{N}\}$ is bounded above. By the *least upper bound property*, there exists $y = \sup X \geq 0$.

First, we will show that $y < 1$. By the assumption, there exists $m \in \mathbb{N}$ such that $a_m \neq b - 1$. Then, for all $n > m$, we have

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ \implies x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}. \end{aligned}$$

Since $a_m \leq b - 2$, we have

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b-1}{b^k} = 1 - \frac{1}{b^m}. \end{aligned}$$

We have $x_n \leq x_m$ for all $n \leq m$, and $x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$ for all $n > m$.

Thus $1 - \frac{1}{b^m}$ is an upper bound of X . Since $y = \sup X$, we have

$$y \leq 1 - \frac{1}{b^m} < 1 \implies y \in [0, 1).$$

Next, we will show that $(a_n)_{n=1}^\infty$ is the *greedy base b expansion* of y . Let $n \in \mathbb{N}$ be fixed. By the assumption, there exists $m \in \mathbb{N}$ such that $m \geq n$ and $a_m \leq b - 2$.

$$\begin{aligned} x_m &= x_n + \sum_{k=n+1}^m \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^m \frac{b-1}{b^k} \\ \implies x_m &< x_n + \frac{1}{b^n} - \frac{1}{b^m} \implies x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}. \end{aligned}$$

Since $a_m \leq b - 2$, $x_m + \frac{1}{b^m}$ is an upper bound of X by the same argument as above. Since $y = \sup X$, we have

$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \implies 0 \leq y - x_n < \frac{1}{b^n}.$$

Therefore, $(a_n)_{n=1}^\infty$ is the *greedy base b expansion* of y . □

Theorem 1.5.11. *Let $b \in \mathbb{N}$ be given with $b \geq 2$ and let $(a_n)_{n=1}^\infty$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists some $N \in \mathbb{N}$ such that*

$$a_n = b - 1 \quad \text{for all } n \geq N,$$

then there exists no $y \in [0, 1)$ such that $(a_n)_{n=1}^\infty$ is the greedy base b expansion of y . This theorem is the converse of [Theorem 1.5.10](#).

Proof. Assume that there exists some $y \in [0, 1)$ such that $(a_n)_{n=1}^\infty$ is the *greedy base b expansion* of y . Then we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \quad \text{for all } n \in \mathbb{N}.$$

Let $N \in \mathbb{N}$ be such that $a_n = b - 1$ for all $n \geq N$. Let $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Then, for all $n \geq N$, we have

$$\begin{aligned} x_n &= x_{N-1} + \sum_{k=N}^n \frac{b-1}{b^k} = x_{N-1} + \frac{1}{b^{N-1}} - \frac{1}{b^n} \\ \implies y - x_n &= y - x_{N-1} - \frac{1}{b^{N-1}} + \frac{1}{b^n} \geq 0 \\ \implies \frac{1}{b^n} &\geq x_{N-1} + \frac{1}{b^{N-1}} - y. \end{aligned}$$

Since $0 \leq y - x_{N-1} < \frac{1}{b^{N-1}}$, we have $x_{N-1} + \frac{1}{b^{N-1}} - y > 0$. By [Corollary 1.5.2](#), there exists some $n \in \mathbb{N}$ with $n \geq N$ such that

$$\frac{1}{b^n} < x_{N-1} + \frac{1}{b^{N-1}} - y,$$

which is a contradiction. Therefore, there exists no $y \in [0, 1)$ such that $(a_n)_{n=1}^\infty$ is the *greedy base b expansion* of y . $\square$

1.6 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on $\overline{\mathbb{R}}$:

- $x + (+\infty) := +\infty$ for all $x \in \mathbb{R} \cup \{+\infty\}$.
- $x + (-\infty) := -\infty$ for all $x \in \mathbb{R} \cup \{-\infty\}$.
- $x \cdot (\pm\infty) := \pm\infty$ for all $x \in \mathbb{R}^+ \cup \{+\infty\}$.
- $x \cdot (\pm\infty) := \mp\infty$ for all $x \in \mathbb{R}^- \cup \{-\infty\}$.
- $\frac{x}{\pm\infty} := 0$ for all $x \in \mathbb{R}$.

The following operations are undefined in $\overline{\mathbb{R}}$:

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\frac{(\pm\infty)}{(\pm\infty)}$ and $\frac{(\pm\infty)}{(\mp\infty)}$
- $\frac{x}{0}$ for all $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

Remark. We can identify the following intervals in $\overline{\mathbb{R}}$:

- $(-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$ for all $a \in \mathbb{R}$.
- $(a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$ for all $a \in \mathbb{R}$.
- $(-\infty, +\infty) = \mathbb{R}$.

Theorem 1.6.1. Any subset of $\overline{\mathbb{R}}$ has a least upper bound and a greatest lower bound in $\overline{\mathbb{R}}$.

Proof. Let $E \subseteq \overline{\mathbb{R}}$. If $E = \emptyset$, then we have

- $\sup E = -\infty$
- $\inf E = +\infty$.

Otherwise, we have the following cases:

- If E is bounded above in $\mathbb{R}$, then $\sup E$ exists in $\mathbb{R}$ by the *least upper bound property* of $\mathbb{R}$.
- If E is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$.
- If E is bounded below in $\mathbb{R}$, then $\inf E$ exists in $\mathbb{R}$ by [Theorem 1.3.7](#).
- If E is not bounded below in $\mathbb{R}$, then $\inf E = -\infty$.

□

1.7 The Complex Field

The complex field $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on $\mathbb{C}$:

- $(x, y) + (u, v) := (x + u, y + v)$ for all $(x, y), (u, v) \in \mathbb{C}$.
- $(x, y) \cdot (u, v) := (xu - yv, xv + yu)$ for all $(x, y), (u, v) \in \mathbb{C}$.

$\mathbb{C}$ is a field with the following properties:

- The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.
- The additive inverse of (x, y) is $(-x, -y)$ and the multiplicative inverse of (x, y) is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for all $(x, y) \in \mathbb{C}$ with $(x, y) \neq (0, 0)$.

Remark. The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \rightarrow \mathbb{R}$$

defined by $f((x, 0)) = x$ is an *isomorphism*. We can identify $\mathbb{R}$ with the subfield $\{(x, 0) \mid x \in \mathbb{R}\}$ of $\mathbb{C}$. We denote $(x, 0) \in \mathbb{C}$ by $x \in \mathbb{R}$.

Remark. We define $i := (0, 1) \in \mathbb{C}$. Then we have $i^2 = -1$, and any complex number $(x, y) \in \mathbb{C}$ can be written as

$$(x, y) = x + yi.$$

Definition 1.7.1. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. The **real part** of z is defined as

$$\Re(z) := x \text{ or } \operatorname{Re}(z) := x,$$

the **imaginary part** of z is defined as

$$\Im(z) := y \text{ or } \operatorname{Im}(z) := y,$$

and the **complex conjugate** of z is defined as

$$\bar{z} := x - yi.$$

Theorem 1.7.2. Let $z = x + yi \in \mathbb{C}$ and let $w = u + vi \in \mathbb{C}$ for some $x, y, u, v \in \mathbb{R}$. Then we have the following properties:

- $\overline{(\overline{z})} = z$.
- $z = \overline{z} \iff z \in \mathbb{R}$.
- $\overline{z + w} = \overline{z} + \overline{w}$.
- $\overline{zw} = \overline{z} \cdot \overline{w}$.
- $z + \overline{z} = 2\operatorname{Re}(z)$ and $z - \overline{z} = 2i\operatorname{Im}(z)$.
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$.

Proof. The proof follows directly from the definitions and properties of the operations on $\mathbb{C}$. $\square$

Definition 1.7.3. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. the **modulus** of z is defined as

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of $|z|$ follows from [Theorems 1.5.6](#) and [1.7.2](#).

Theorem 1.7.4. Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ and $b_1, b_2, \dots, b_n \in \mathbb{C}$. Then we have

$$\left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2.$$

This theorem is called the **Cauchy-Schwarz inequality**.

Proof. Let $i, j \in \mathbb{N}$ with $1 \leq i < j \leq n$. Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned}
 \sum_{i < j} (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) &= \sum_{i \neq j} (a_i \bar{b}_i \cdot \bar{a}_j b_j) \\
 &= \sum_{i=1}^n a_i \bar{b}_i \cdot \sum_{j=1}^n \bar{a}_j b_j - \sum_{k=1}^n a_k \bar{b}_k \cdot \bar{a}_k b_k \\
 &= \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2.
 \end{aligned}$$

Therefore, we have

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \geq 0.$$

□

Corollary 1.7.5. *Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$. Then we have*

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2.$$

Theorem 1.7.6. *Let $z, w \in \mathbb{C}$. Then we have the following properties:*

- (A) $|zw| = |z| \cdot |w|$.
- (B) $|z + w| \leq |z| + |w|$.
- (C) $||z| - |w|| \leq |z - w|$.

Proof. Let $z = x + yi$ and $w = u + vi$ for some $x, y, u, v \in \mathbb{R}$.

- (A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$.
- (B) By [Corollary 1.7.5](#), we have $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$.

$$\begin{aligned}
 |z + w|^2 &= (x + u)^2 + (y + v)^2 \\
 &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\
 &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
 &= (|z| + |w|)^2.
 \end{aligned}$$

- (C) Again, by [Corollary 1.7.5](#), we have

$$\begin{aligned}
 |z - w|^2 &= (x - u)^2 + (y - v)^2 \\
 &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\
 &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
 &= (|z| - |w|)^2.
 \end{aligned}$$

□

1.8 Euclidean Spaces

For each $n \in \mathbb{N}$, we call the set of all ordered n -tuples of real numbers as the **Euclidean n -space**, and denote it as $\mathbb{R}^n$. We can identify $\mathbb{R}^n$ as

$$\mathbb{R}^n = \{\mathbf{x} = (x_1, x_2, \dots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \dots, n\},$$

where $x_1, x_2, \dots, x_n$ are called the **coordinates** of $\mathbf{x}$. The elements of $\mathbb{R}^n$ with $n > 1$ are called **points** (or *vectors*), which are denoted by boldface letters.

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$, and let $\alpha \in \mathbb{R}$. We define the following operations on $\mathbb{R}^n$:

- $\mathbf{x} + \mathbf{y} := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$.
- $\alpha \mathbf{x} := (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$.

Then, $\mathbb{R}^n$ is a *vector space* over $\mathbb{R}$ with the zero vector $\mathbf{0} = (0, 0, \dots, 0)$.

Definition 1.8.1. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. The **inner product** of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \cdot \mathbf{y} := \sum_{k=1}^n x_k y_k.$$

The **norm** of $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}.$$

The **distance** between $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

We can confirm that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.8.2. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. Then we have the following properties:

- (A) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$.
- (B) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.
- (C) $\|\mathbf{x} - \mathbf{y}\| \geq \left| \|\mathbf{x}\| - \|\mathbf{y}\| \right|$.

Proof. (A) By [Corollary 1.7.5](#), we have

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2.$$

(B) By part (A), we have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

(C) By part (B), we have

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|. \\ \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \implies \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

□

Chapter 2

Basic Topology

2.1 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain** X .
- A **codomain** Y .
- A **rule** f which associates each element $x \in X$ to a *unique* element $y \in Y$.

f is said to be a *function* of X into Y , which is denoted by $f : X \rightarrow Y$. For each $x \in X$, the *unique* element $y \in Y$ associated with x is denoted by $f(x)$.

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq X$, we define the **image** of E under f as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f .

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq Y$, we define the **inverse image** of E under f as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X.$$

Definition 2.1.1. Let $f : X \rightarrow Y$ be a function. We define the properties of f as follows:

- f is **injective** (or *one-to-one*) if for any $x_1, x_2 \in X$, we have

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- f is **surjective** (or *onto*) if $f(X) = Y$.
- f is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

Definition 2.1.2. Let $f : X \rightarrow Y$ be a bijective function. Then, for any $y \in Y$, there exists a *unique* $x \in X$ such that $f(x) = y$. Therefore, we can define a function

$$f^{-1} : Y \rightarrow X \quad \text{such that} \quad f^{-1}(y) = x.$$

Such a function f^{-1} is called the **inverse function** of f .

Theorem 2.1.3. Let $f : X \rightarrow Y$ be a bijective function. Then the inverse function $f^{-1} : Y \rightarrow X$ is bijective.

Proof. Let $y_1, y_2 \in Y$ be given. Assume that $f^{-1}(y_1) = f^{-1}(y_2) = x$. Then, by [Definition 2.1.2](#), we have

$$f(x) = y_1 \text{ and } f(x) = y_2 \implies y_1 = y_2$$

since f is a function. Therefore, we have that f^{-1} is injective. By the definition of image, we have that $f^{-1}(Y) = X$, which implies that f^{-1} is surjective. $\square$

Definition 2.1.4. Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. We define the restriction functions as follows:

- The **restriction** of f to E is defined by

$$f|_E : E \rightarrow Y \quad \text{such that} \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of f to F is defined by

$$f|_F : X \rightarrow F \quad \text{such that} \quad f|_F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of f to E and F is defined by

$$f|_E^F : E \rightarrow F \quad \text{such that} \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

Theorem 2.1.5. Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. Then we have the following properties:

(A) If f is injective, then $f|_E^F$ is also injective.

(B) If $f(E) = F$, then $f|_E^F$ is surjective.

(C) If f is injective and $f(E) = F$, then $f|_E^F$ is bijective.

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in E$.

(A) Assume that f is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

(B) Assume that $f(E) = F$. Then, for any $y \in F$, there exists $x \in E$ such that $f(x) = y$. Therefore, we have $f|_E^F(E) = F$.

□

Definition 2.1.6. Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. We define the **composition** of f and g as the function $g \circ f : X \rightarrow Z$ defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

Theorem 2.1.7. Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. Then we have the following properties:

(A) If f and g are injective, then $g \circ f$ is also injective.

(B) If f and g are surjective, then $g \circ f$ is also surjective.

(C) If f and g are bijective, then $g \circ f$ is also bijective.

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in X$ be given.

(A) Assume that f and g are injective. Then we have

$$\begin{aligned} (g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ &\implies f(x_1) = f(x_2) \implies x_1 = x_2. \end{aligned}$$

(B) Assume that f and g are surjective. Then, for any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, we have $(g \circ f)(x) = g(f(x)) = g(y) = z$.

□

Theorem 2.1.8. *Let S be a set of sets. For [Sections 2.2](#) and [2.3](#), we consider the relation $\sim$ defined on S by*

$$X \sim Y \iff \text{There exists a bijection } f : X \rightarrow Y.$$

The relation $\sim$ is an equivalence relation on S .

Proof. Let $X, Y, Z \in S$ be given.

- **Reflexivity:** The identity function $id_X : X \rightarrow X$, defined by $id_X(x) = x$ for all $x \in X$, is bijective. Therefore, we have $X \sim X$.
- **Symmetry:** Assume that $X \sim Y$. Then there exists a bijection $f : X \rightarrow Y$. By [Theorem 2.1.3](#), the inverse function $f^{-1} : Y \rightarrow X$ is also bijective. Therefore, we have $Y \sim X$.
- **Transitivity:** Assume that $X \sim Y$ and $Y \sim Z$. Then there exist bijections $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. By [Theorem 2.1.7](#), the composition $g \circ f : X \rightarrow Z$ is also bijective. Therefore, we have $X \sim Z$.

□

2.2 Finite Sets

Let J_n be a set defined by

$$J_n := \{1, 2, \dots, n\} \quad \text{for each } n \in \mathbb{N}.$$

A set X is said to be **finite** if $J_n \sim X$ for some $n \in \mathbb{N}$. In this case, n is called the **cardinality** of X , denoted by $|X|$. Especially, if $X = \emptyset$, we define $|X| = 0$. X is said to be **infinite** if X is not finite.

Lemma 2.2.1. *Any subset of J_n is finite for each $n \in \mathbb{N}$.*

Proof. Let $P(n)$ be the statement that any subset of J_n is finite.

- **Base case:** $J_1 = \{1\}$ has $\emptyset$ and J_1 as its only subsets. We have $J_1 \sim J_1$ by $f \equiv id_{J_1}$, and $\emptyset$ is finite by definition. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $A \subseteq J_{n+1}$ be given. We have two cases:
 - If $n+1 \notin A$, then $A \subseteq J_n$, which is finite by the inductive hypothesis.
 - If $n+1 \in A$, then $A = B \cup \{n+1\}$ for some $B \subseteq J_n$. By the inductive hypothesis, B is finite, so there exists a bijection $f : J_m \rightarrow B$ for some $m \in \mathbb{N}$. Let $g : J_{m+1} \rightarrow A$ be defined by

$$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n+1 & \text{if } k = m+1 \end{cases}$$

Then g is bijective, which implies that A is finite.

Thus $P(n + 1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.2.2. *Any subset of a finite set is finite.*

Proof. Let X be a finite set and $Y \subseteq X$. Then there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. Let Z be defined by

$$Z := f^{-1}(Y) \subseteq J_n.$$

Since $Z \subseteq J_n$, Z is finite by Lemma 2.2.1, so there exists a bijection $g : J_m \rightarrow Z$ for some $m \in \mathbb{N}$.

By Theorem 2.1.5, $f|_Z^Y$ is bijective. Then, by Theorem 2.1.7, the composition $f|_Z^Y \circ g : J_m \rightarrow Y$ is also bijective. Therefore, Y is finite. $\square$

Lemma 2.2.3. *Let J_n be given for some $n \in \mathbb{N}$. If there exists an surjection $f : J_n \rightarrow Y$, then Y is finite.*

Proof. Let $f : J_n \rightarrow Y$ be a surjective function. For any $y \in Y$, let N_y be a set defined by

$$N_y := f^{-1}(\{y\}) \subseteq J_n.$$

Since f is surjective, N_y is nonempty. By Theorem 1.1.3, N_y has a least element, say n_y . Let M_Y be a set defined by

$$M_Y := \{n_y \mid y \in Y\} \subseteq J_n.$$

Let $g : M_Y \rightarrow Y$ be a function defined by

$$g(n_y) = y \quad \text{for each } n_y \in M_Y.$$

Then g is bijective. By Lemma 2.2.1, M_Y is finite, so there exists a bijection $h : J_m \rightarrow M_Y$ for some $m \in \mathbb{N}$. Then, by Theorem 2.1.7, the composition $g \circ h : J_m \rightarrow Y$ is bijective. Therefore, Y is finite. $\square$

Theorem 2.2.4. *Let X be a finite set. If there exists an surjection $f : X \rightarrow Y$, then Y is finite.*

Proof. Let $f : X \rightarrow Y$ be a surjective function. Since X is finite, there exists a bijection $g : J_n \rightarrow X$ for some $n \in \mathbb{N}$. Then, by Theorem 2.1.7, the composition $f \circ g : J_n \rightarrow Y$ is surjective. Therefore, by Lemma 2.2.3, Y is finite. $\square$

Theorem 2.2.5. *Let X be a finite nonempty ordered set. Then X has both a least element and a greatest element.*

Proof. Since X is finite and nonempty, there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. The elements of X can be listed as $X = \{a_1, a_2, \dots, a_n\}$, where $a_k = f(k)$ for each $k = 1, 2, \dots, n$.

First, we will show that X has a least element. Let $P(n)$ be the statement that any finite ordered set of cardinality n has a least element.

- **Base case:** $X = \{a_1\}$ has a least element a_1 . Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n + 1$ such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a least element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:

- If $a_k \leq a_{n+1}$, then a_k is the least element of X .
- If $a_{n+1} < a_k$, then a_{n+1} is the least element of X .

Thus $P(n + 1)$ is true whenever $P(n)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$.

Next, we will show that X has a greatest element. Let $Q(n)$ be the statement that any finite ordered set of cardinality n has a greatest element.

- **Base case:** $X = \{a_1\}$ has a greatest element a_1 . Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n + 1$ such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a greatest element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:

- If $a_k \geq a_{n+1}$, then a_k is the greatest element of X .
- If $a_{n+1} > a_k$, then a_{n+1} is the greatest element of X .

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. $\square$

Lemma 2.2.6. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

- (A) *If there exists an injection $f : J_m \rightarrow J_n$, then we have $m \leq n$.*
- (B) *If there exists a surjection $f : J_m \rightarrow J_n$, then we have $m \geq n$.*
- (C) *If there exists a bijection $f : J_m \rightarrow J_n$, then we have $m = n$.*

This lemma guarantees the uniqueness of the cardinality. If $J_m \sim X$ and $J_n \sim X$, then we have $J_m \sim J_n$, which implies that $m = n$ by (C).

Proof. We will prove (A) by mathematical induction on m , (B) by mathematical induction on n , and (C) follows from (A) and (B).

(A) Let $n \in \mathbb{N}$ be fixed. Let $P(m)$ be the statement that if there exists an injective function $f_m : J_m \rightarrow J_n$, then we have $m \leq n$.

- **Base case:** Let $m = 1$. Then $m \leq n$ holds. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(m)$ is true for some $m \in \mathbb{N}$. If $n = 1$, there does not exist an injective function $f_{m+1} : J_{m+1} \rightarrow J_n$ since $f_{m+1}(1) \neq f_{m+1}(m+1)$, where $P(m+1)$ is true.

Otherwise, assume that such a function f_{m+1} exists. We have two cases:

- If $f_{m+1}(m+1) = n$, let f_m be a function defined by

$$f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}} : J_m \rightarrow J_{n-1}.$$

Then f_m is injective by [Theorem 2.1.5](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

- If $f_{m+1}(m+1) = x \neq n$, let $g_n : J_n \rightarrow J_n$ be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then g_n is bijective. Let f_m be a function defined by

$$f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}} : J_m \rightarrow J_{n-1}.$$

Then f_m is injective by [Theorems 2.1.5](#) and [2.1.7](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

Therefore, $P(m+1)$ is true whenever $P(m)$ is true.

By the *principle of mathematical induction*, $P(m)$ is true for all $m \in \mathbb{N}$.

(B) Let $m \in \mathbb{N}$ be fixed. Let $Q(n)$ be the statement that if there exists a surjective function $f_n : J_m \rightarrow J_n$, then we have $m \geq n$.

- **Base case:** Let $n = 1$. Then $m \geq n$ holds. Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. If $m = 1$, there does not exist a surjective function $f_{n+1} : J_m \rightarrow J_{n+1}$ since $f_{n+1}(1)$ is unique, where $Q(n+1)$ is true.

Otherwise, assume that such a function f_{n+1} exists. We have two cases:

- If $f_{n+1}(m) = n+1$, let K_{n+1} be a set defined by

$$K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let $f_n : J_m \rightarrow J_n$ be a function defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then f_n is surjective. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

- If $f_{n+1}(m) = x \neq n+1$, let $g_{n+1} : J_{n+1} \rightarrow J_{n+1}$ be a function defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then g_{n+1} is bijective. Let K_{n+1} be a set defined by

$$K_{n+1} := (g_{n+1} \circ f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let $f_n : J_m \rightarrow J_n$ be a function defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then f_n is surjective by [Theorem 2.1.7](#). By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the *principle of mathematical induction*, $Q(n)$ is true for all $n \in \mathbb{N}$.

□

Lemma 2.2.7. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

(A) *If $m \leq n$, then there exists an injection $f : J_m \rightarrow J_n$.*

(B) *If $m \geq n$, then there exists a surjection $f : J_m \rightarrow J_n$.*

(C) *If $m = n$, then there exists a bijection $f : J_m \rightarrow J_n$.*

This lemma is the converse of [Lemma 2.2.6](#).

Proof. We will prove this lemma by explicitly constructing the functions.

(A) If $m \leq n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is an injection.

(B) If $m \geq n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = \begin{cases} i, & \text{if } i \leq n \\ n, & \text{if } i > n \end{cases}$$

Then f is a surjection.

(C) If $m = n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is a bijection.

□

Theorem 2.2.8. *Let X, Y be finite sets. Then we have the following properties:*

(A) *If there exists an injection $f : X \rightarrow Y$, then we have $|X| \leq |Y|$.*

(B) *If there exists a surjection $f : X \rightarrow Y$, then we have $|X| \geq |Y|$.*

(C) *If there exists a bijection $f : X \rightarrow Y$, then we have $|X| = |Y|$.*

Moreover, the converses of (A), (B), and (C) are also true.

*The contraposition of (A) is called the **pigeonhole principle**, which states that if $|X| > |Y|$, then there exists no injection $f : X \rightarrow Y$.*

Proof. Let $|X| = m$ and $|Y| = n$ for some $m, n \in \mathbb{N}$. Then there exist bijections $f_X : J_m \rightarrow X$ and $f_Y : J_n \rightarrow Y$. By [Theorems 2.1.3](#) and [2.1.7](#), the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \rightarrow J_n$$

is injective, surjective, or bijective if f is injective, surjective, or bijective, respectively. Then, by [Lemma 2.2.6](#), we have the desired results:

- (A) If f is injective, then F is injective, which implies that $m \leq n$.
- (B) If f is surjective, then F is surjective, which implies that $m \geq n$.
- (C) If f is bijective, then F is bijective, which implies that $m = n$.

Also, by [Theorems 2.1.3](#) and [2.1.7](#), the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \rightarrow Y$$

is injective, surjective, or bijective if $G : J_m \rightarrow J_n$ is injective, surjective, or bijective, respectively. Then, by [Lemma 2.2.7](#), we have the desired converses:

- (A) If $m \leq n$, then there exists an injective function $G : J_m \rightarrow J_n$, which implies that there exists an injective function $f : X \rightarrow Y$.
- (B) If $m \geq n$, then there exists a surjective function $G : J_m \rightarrow J_n$, which implies that there exists a surjective function $f : X \rightarrow Y$.
- (C) If $m = n$, then there exists a bijective function $G : J_m \rightarrow J_n$, which implies that there exists a bijective function $f : X \rightarrow Y$.

□

Lemma 2.2.9. *Let X, Y be finite sets with $X \cap Y = \emptyset$. Then the union $X \cup Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \rightarrow X$ and $g : J_n \rightarrow Y$. Let $h : J_{m+n} \rightarrow X \cup Y$ be a function defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i - m), & \text{if } m < i \leq m + n \end{cases}$$

Then h is bijective, which implies that $X \cup Y$ is finite.

□

Theorem 2.2.10. *Let X, Y be finite sets. Then the union $X \cup Y$ is finite.*

Proof. Let A, B, C be sets defined by

- $A = X \setminus Y \subseteq X$.
- $B = Y \setminus X \subseteq Y$.
- $C = X \cap Y \subseteq X$.

Then A, B, C are finite by [Theorem 2.2.2](#) and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$.
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$.
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$.

By [Lemma 2.2.9](#), $X \cup Y$ is finite. □

Theorem 2.2.11. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the union $\bigcup_{k=1}^n X_k$ is finite, i.e., any finite union of finite sets is finite.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \bigcup_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then $\bigcup_{k=1}^1 X_k = X_1$ is finite. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have

$$\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^n X_k \right) \cup X_{n+1}.$$

By the inductive hypothesis, $\bigcup_{k=1}^n X_k$ is finite. Then, by [Theorem 2.2.10](#),

$\bigcup_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.2.12. *Let X, Y be finite sets. Then the cartesian product $X \times Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \rightarrow X$ and $g : J_n \rightarrow Y$. Let $h : J_{mn} \rightarrow X \times Y$ be a function defined by

$$h(i) = \left(f\left(\left[\frac{i}{n}\right]\right), g\left(i - n\left(\left[\frac{i}{n}\right] - 1\right)\right) \right) \text{ for all } i \in J_{mn}.$$

Then h is bijective, which implies that $X \times Y$ is finite. $\square$

Theorem 2.2.13. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the cartesian product $\prod_{k=1}^n X_k$ is finite.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \prod_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then, $\prod_{k=1}^1 X_k = X_1$ is finite. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have

$$\prod_{k=1}^{n+1} X_k = \left(\prod_{k=1}^n X_k \right) \times X_{n+1}.$$

By the inductive hypothesis, $\prod_{k=1}^n X_k$ is finite. Then, by [Lemma 2.2.12](#), $\prod_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.2.14. *Let X be a nonempty set. Then there exists no surjection $f : X \rightarrow \mathcal{P}(X)$, where*

$$\mathcal{P}(X) := \{E \mid E \subseteq X\}$$

*is called the **power set** of X . This theorem is called **Cantor's theorem**.*

Proof. Let S be a set defined by

$$S := \{x \in X \mid x \notin f(x)\}.$$

Since $S \subseteq X$, we have that $S \in \mathcal{P}(X)$. Assume that there exists a surjection $f : X \rightarrow \mathcal{P}(X)$. Then there exists $x \in X$ such that $f(x) = S$.

If $x \in S$, then we have $x \notin f(x) = S$, which is a contradiction. If $x \notin S$, then we have $x \in f(x) = S$, which is also a contradiction. Therefore, there exists no surjection $f : X \rightarrow \mathcal{P}(X)$. $\square$

2.3 Countable and Uncountable Sets

A set X is said to be **countable** if $\mathbb{N} \sim X$. A set X is said to be **uncountable** if it is neither finite nor countable. A set X is said to be **at most countable** if X is either finite or countable.

Theorem 2.3.1. *Let X be an infinite set, and let Y be a finite set. Then $X \setminus Y$ is infinite.*

Proof. Assume that $X \setminus Y$ is finite. Since Y is finite, by [Theorem 2.2.10](#), $(X \setminus Y) \cup Y$ is finite. Then, by [Theorem 2.2.2](#), $X \subseteq (X \setminus Y) \cup Y$ is finite, which contradicts the assumption. Therefore, $X \setminus Y$ is infinite. $\square$

Theorem 2.3.2. *Let X be an infinite set. If there exists an injection $f : X \rightarrow Y$, then Y is infinite.*

Proof. Assume that Y is finite. Let $g : Y \rightarrow X$ be defined by

$$g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{Otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. Then g is surjective. By [Theorem 2.2.4](#), X is finite, which is a contradiction. Therefore, Y is infinite. $\square$

Lemma 2.3.3. $\mathbb{N}$ is infinite.

Proof. Assume that $\mathbb{N}$ is finite. Then there exists a bijection $f : J_n \rightarrow \mathbb{N}$ for some $n \in \mathbb{N}$. Then, by [Theorem 2.2.2](#), $f(J_n) \subseteq \mathbb{N}$ is finite. By [Theorem 2.2.5](#), $f(J_n)$ has a greatest element, say m .

Then we have $m+1 \in \mathbb{N}$ and $m < m+1 \notin f(J_n)$, which contradicts the assumption that f is surjective. Therefore, $\mathbb{N}$ is infinite. $\square$

Theorem 2.3.4. *Any countable set is infinite.*

Proof. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. By [Theorem 2.3.2](#) and [lemma 2.3.3](#), X is infinite. $\square$

Remark. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. The elements of X can be listed as $X = \{x_1, x_2, x_3, \dots\}$, where $x_n = f(n)$ for all $n \in \mathbb{N}$. In other words, the elements of X can be arranged in a sequence.

Theorem 2.3.5. *$\mathbb{Z}$ is countable.*

Proof. Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be a function defined by

$$f(n) = \begin{cases} -\frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that f is bijective, which implies that $\mathbb{Z}$ is countable.

- **Injectivity:** Assume that $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{N}$. Since $f(n) \geq 0$ for odd n and $f(n) < 0$ for even n , we have n_1 and n_2 are either both even or both odd.

- If both n_1 and n_2 are even, then we have $-\frac{n_1}{2} = -\frac{n_2}{2}$, which implies $n_1 = n_2$.
- If both n_1 and n_2 are odd, then we have $\frac{n_1-1}{2} = \frac{n_2-1}{2}$, which implies $n_1 = n_2$.

Thus f is injective.

- **Surjectivity:** Let $k \in \mathbb{Z}$ be given. We need to find an $n \in \mathbb{N}$ such that $f(n) = k$.

- If $k \geq 0$, let $n = 2k + 1$. Then we have $f(n) = \frac{n-1}{2} = \frac{2k}{2} = k$.
- If $k < 0$, let $n = -2k$. Then we have $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$.

Thus f is surjective.

$\square$

Lemma 2.3.6. *Any infinite subset of $\mathbb{N}$ is countable.*

Proof. Let $X \subseteq \mathbb{N}$ be an infinite subset. Let $(X_n)_{n=1}^\infty$ be a sequence and let $f : \mathbb{N} \rightarrow X$ be a function defined as follows:

- $X_1 = X, f(1) = \min X_1$
- $X_2 = X_1 \setminus \{f(1)\}, f(2) = \min X_2$
- $\vdots$
- $X_{n+1} = X_n \setminus \{f(n)\}, f(n+1) = \min X_{n+1}$

for each $n \in \mathbb{N}$. By [Theorem 2.3.1](#), X_n is infinite for each $n \in \mathbb{N}$, so is nonempty. Then, by [Theorem 1.1.3](#), X_n has a least element, so f is well-defined. We will show that f is bijective, which implies that X is countable.

- **Injectivity:** Assume that $n_1 \neq n_2$ for some $n_1, n_2 \in \mathbb{N}$. Without loss of generality, assume that $n_1 < n_2$. Then we have

$$f(n_1) \in X_{n_1} \text{ and } f(n_2) \in X_{n_2}.$$

Since $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$, we have $f(n_2) \neq f(n_1)$. Thus f is injective.

- **Surjectivity:** Let $x \in X$ be given. Let S be a set defined as

$$S := \{n \in X \mid n < x\}.$$

If $S = \emptyset$, then $f(1) = x$. Otherwise, by [Lemma 2.2.1](#), $S \subseteq J_{x-1}$ is finite. Let $k = |S|$, $S = \{s_1, s_2, \dots, s_k\}$ with $s_1 < s_2 < \dots < s_k$. Then we have

- $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}.$
- $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}.$
- $\vdots$
- $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S.$

Thus $f(k+1) = \min X_{k+1} = x$. Therefore, f is surjective.

□

Theorem 2.3.7. *Any infinite subset of a countable set is countable.*

Proof. Let X be a countable set and let $Y \subseteq X$ be an infinite subset. Since X is countable, there exists a bijection $f : \mathbb{N} \rightarrow X$. By [Theorem 2.1.5](#), $f|_{f^{-1}(Y)}^Y : f^{-1}(Y) \rightarrow Y$ is bijective. Then, by [Theorem 2.2.4](#), $f^{-1}(Y) \subseteq \mathbb{N}$ is infinite.

By [Lemma 2.3.6](#), $f^{-1}(Y)$ is countable. Then there exists a bijection $g : \mathbb{N} \rightarrow f^{-1}(Y)$. The composition $f|_{f^{-1}(Y)}^Y \circ g : \mathbb{N} \rightarrow Y$ is bijective, which implies that Y is countable. □

Theorem 2.3.8. *Let X be infinite. If there exists a surjection $f : \mathbb{N} \rightarrow X$, then X is countable.*

Proof. For each $k \in X$, let S_k be a set defined as

$$S_k := f^{-1}(\{k\}).$$

Since f is surjective, $S_k \neq \emptyset$ and by Theorem 1.1.3, S_k has a least element for all $k \in X$. Let $g : X \rightarrow \mathbb{N}$ be a function defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Assume that $g(k_1) = g(k_2)$ for some $k_1, k_2 \in X$, i.e., $\min S_{k_1} = \min S_{k_2}$. Since $S_i \cap S_j = \emptyset$ for all $i, j \in X$ with $i \neq j$, we have $S_{k_1} = S_{k_2} \implies k_1 = k_2$. Thus g is injective.

Since g is injective, by Theorem 2.1.5, $g|^{g(X)} : X \rightarrow g(X)$ is bijective. Since X is infinite, $g(X) \subseteq \mathbb{N}$ is infinite by Theorem 2.3.2. Then, by Lemma 2.3.6, $g(X)$ is countable, which implies that X is countable. $\square$

Theorem 2.3.9. *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n=1}^{\infty} X_n$ is countable, i.e., any countable union of countable sets is countable.*

Proof. There exists a bijection $f_n : \mathbb{N} \rightarrow X_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define the followings:

- $t_n := \min T_n$, where $T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$.
- $s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$.

Since $2n \in T_n$, T_n is nonempty. Then, by Theorem 1.1.3, T_n has a least element, so t_n is well-defined. Since $\frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2}$, we have

$$t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1,$$

so $t_n - s_n + 1 \in \mathbb{N}$. Let $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ be a function defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then there exists some $m \in \mathbb{N}$ such that $x \in X_m$. Since f_m is surjective, there exists some $l \in \mathbb{N}$ such that $f_m(l) = x$. Let

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}.$$

We will show that $l+m-2 \notin T_k$ and $l+m-1 \in T_k$, which implies that

- $t_k = \min T_k = l+m-1$.
- $s_k = k - \frac{t_k(t_k-1)}{2} = m$.
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l+m-1-m+1) = f_m(l) = x$.

Consider the following inequalities when $z = l+m-2$ and $z = l+m-1$:

- Let $z = l+m-2$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let $z = l+m-1$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1-l \leq 0. \end{aligned}$$

Thus g is surjective, which implies that $\bigcup_{n=1}^{\infty} X_n$ is countable by [Theorem 2.3.8](#). $\square$

Corollary 2.3.10. *Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then*

$\bigcup_{i=1}^n X_i$ is countable, i.e., any finite union of countable sets is countable.

Proof. Let $X_{n+1}, X_{n+2}, \dots$ be countable sets. Then, by [Theorem 2.3.9](#), $\bigcup_{n=1}^{\infty} X_n$ is countable. Assume that $\bigcup_{i=1}^n X_i$ is finite. Then $X_1 \subseteq \bigcup_{i=1}^n X_i$ is finite by [Theorem 2.2.2](#), which contradicts [Theorem 2.3.4](#). Thus $\bigcup_{i=1}^n X_i$ is infinite. Since $\bigcup_{i=1}^n X_i \subseteq \bigcup_{n=1}^{\infty} X_n$, by [Theorem 2.3.7](#), $\bigcup_{i=1}^n X_i$ is countable. $\square$

Theorem 2.3.11. *Let X be a countable set. If Y_x is finite for each $x \in X$, then the union*

$$Y = \bigcup_{x \in X} Y_x$$

is countable, i.e., any countable union of finite sets is countable.

Proof. There exists a bijection $f : \mathbb{N} \rightarrow X$. Also, for each $x \in X$, there exists a bijection $g_x : J_{n_x} \rightarrow Y_x$ for some $n_x \in \mathbb{N}$. Let $h : \mathbb{N} \rightarrow Y$ be a function defined by

- $h(j) = g_{f(1)}(j)$ for $j = 1, 2, \dots, n_{f(1)}$.
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$ for each $k \in \mathbb{N}$ with $k \geq 2$, and $m = 1, 2, \dots, n_{f(k)}$.

Then h is surjective. Therefore, by [Theorem 2.3.8](#), Y is countable. $\square$

Lemma 2.3.12. $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. We will prove this lemma by explicitly constructing a bijection from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be a function defined by

$$f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

for each $(m, n) \in \mathbb{N} \times \mathbb{N}$. We will show that f is bijective.

- **Injectivity:** Assume that $(m_1, n_1) \neq (m_2, n_2)$. If $m_1 + n_1 = m_2 + n_2$, then $m_1 \neq m_2$, so $f(m_1, n_1) \neq f(m_2, n_2)$. Otherwise, $m_1 + n_1 \neq m_2 + n_2$.

Without loss of generality, assume $m_1 + n_1 < m_2 + n_2$. Then, since $m_1 +$

$n_1 + 1 \leq m_2 + n_2$, we have

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2), \end{aligned}$$

Thus $f(m_1, n_1) \neq f(m_2, n_2)$, so f is injective.

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Let t_k, s_k be defined as in the proof of [Theorem 2.3.9](#). Then we have $f(s_k, t_k - s_k + 1) = k$. Thus f is surjective.

□

Theorem 2.3.13. *Let X and Y be countable sets. Then the cartesian product $X \times Y$ is countable.*

Proof. There exist bijections $f : \mathbb{N} \rightarrow X$ and $g : \mathbb{N} \rightarrow Y$. Let $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$ be a function defined by

$$h(m, n) = (f(m), g(n)) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}.$$

Then h is bijective. By [Lemma 2.3.12](#), $X \times Y$ is countable.

□

Theorem 2.3.14. *Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\prod_{i=1}^n X_i$ is countable, i.e., any finite product of countable sets is countable.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are countable sets} \implies \prod_{i=1}^n X_i \text{ is countable.}$$

- **Base case:** Let X_1 be a countable set. Then $\prod_{i=1}^1 X_i = X_1$ is countable. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be countable sets. Then we have

$$\prod_{i=1}^{n+1} X_i = \left(\prod_{i=1}^n X_i \right) \times X_{n+1}$$

By the inductive hypothesis, $\prod_{i=1}^n X_i$ is countable. Then, by [Theorem 2.3.13](#),

$\prod_{i=1}^{n+1} X_i$ is countable. Thus $P(n+1)$ is true when $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.15. $\mathbb{Q}$ is countable.

Proof. Any $q \in \mathbb{Q}$ can be expressed as $q = \frac{m}{n}$ for some $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Let $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ be a function defined by

$$f(m, n) = \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

Then f is surjective. By [Theorems 2.3.5](#) and [2.3.13](#), $\mathbb{Z} \times \mathbb{N}$ is countable. Then there exists a bijection $g : \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{N}$.

By [Theorem 2.1.7](#), $f \circ g : \mathbb{N} \rightarrow \mathbb{Q}$ is surjective. Since $\mathbb{N} \subseteq \mathbb{Q}$, by [Lemma 2.3.3](#) and [theorem 2.2.2](#), $\mathbb{Q}$ is infinite. Therefore, by [Theorem 2.3.8](#), $\mathbb{Q}$ is countable. $\square$

Lemma 2.3.16. $X := \prod_{n=1}^{\infty} \{0, 1\}$ is uncountable.

Proof. For each $n \in \mathbb{N}$, let $e_n \in X$ be a sequence defined by

$$e_{nm} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{Otherwise} \end{cases}$$

Let $f : \mathbb{N} \rightarrow X$ be a function defined by

$$f(n) = e_n \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By [Lemma 2.3.3](#) and [theorem 2.3.2](#), X is infinite.

Assume that X is countable. Then, there exists a bijection $g : \mathbb{N} \rightarrow X$. Let $X = \{g(1), g(2), g(3), \dots\}$, where

- $g(1) = (a_{11}, a_{12}, a_{13}, \dots)$.
- $g(2) = (a_{21}, a_{22}, a_{23}, \dots)$.
- $\vdots$

Let $x = (b_1, b_2, b_3, \dots) \in X$ be a sequence defined by

$$b_n = 1 - a_{nn} \text{ for all } n \in \mathbb{N}.$$

Then, $x \neq g(n)$ for all $n \in \mathbb{N}$, which contradicts the assumption that g is surjective. Therefore, X is uncountable. $\square$

Theorem 2.3.17. $\mathbb{R}$ is uncountable.

Proof. Since $\mathbb{N} \subseteq \mathbb{R}$, by Lemma 2.3.3 and theorem 2.2.2, $\mathbb{R}$ is infinite. Let $f : \mathbb{N} \rightarrow [0, 1)$ be a function defined by

$$f(n) = \frac{1}{n+1} \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By Lemma 2.3.3 and theorem 2.3.2, $[0, 1)$ is infinite. We will prove that $[0, 1) \subseteq \mathbb{R}$ is uncountable, which implies that $\mathbb{R}$ is uncountable by Theorem 2.3.7.

Let $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ be a function defined by

$$g(x) = (a_n^x)_{n=1}^{\infty} \text{ for all } x \in [0, 1)$$

where $(a_n^x)_{n=1}^{\infty}$ is the greedy base 2 expansion of x . Let Y be a set defined by

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}.$$

Then, by Theorems 1.5.9 and 2.1.5, $g|_Y$ is bijective.

Assume that $[0, 1)$ is countable. Then there exists a bijection $h : \mathbb{N} \rightarrow [0, 1)$. By Theorem 2.1.7, $g|_Y \circ h : \mathbb{N} \rightarrow Y$ is bijective, so Y is countable. Let $Z :=$

$\prod_{n=1}^{\infty} \{0, 1\} \setminus Y$. By Theorems 1.5.10 and 1.5.11, we have

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \text{ for some } N \in \mathbb{N} \right\}.$$

Let Z_N be a set defined by

$$Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \right\} \text{ for each } N \in \mathbb{N}.$$

Let $F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$ be a function defined by

$$F_N\left((a_n)_{n=1}^{N-1}\right) = (a_n)_{n=1}^{N-1} \cup (1)_{n=N}^{\infty} \text{ for all } (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0, 1\}.$$

Then F_N is surjective. By [Theorems 2.2.4](#) and [2.2.13](#), Z_N is finite. Then, by [Theorem 2.3.11](#), $Z = \bigcup_{N=1}^{\infty} Z_N$ is countable.

By [Corollary 2.3.10](#), $\prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z$ is countable. [Lemma 2.3.16](#) shows that this is a contradiction. Therefore, $[0, 1)$ is uncountable. $\square$

2.4 Metric Spaces

Let X be a set. A function $d : X \times X \rightarrow \mathbb{R}$ is called a **metric** on X if for any $p, q \in X$, we have

$$\text{(A)} \quad d(p, q) \geq 0 \text{ and } d(p, q) = 0 \iff p = q.$$

$$\text{(B)} \quad d(p, q) = d(q, p).$$

$$\text{(C)} \quad d(p, r) \leq d(p, q) + d(q, r) \text{ for any } r \in X.$$

(C) is called the **triangle inequality**. The pair (X, d) is called a **metric space**.

Theorem 2.4.1. For some $n \in \mathbb{N}$, let $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a function defined by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. Especially, d is called the **standard metric** on $\mathbb{R}^n$.

Proof. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be given.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$ is trivial. Also, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) = 0 &\iff \|\mathbf{x} - \mathbf{y}\| = 0 \\ &\iff x_k = y_k \text{ for all } k = 1, 2, \dots, n \\ &\iff \mathbf{x} = \mathbf{y}. \end{aligned}$$

- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ is trivial.
- By [Theorem 1.8.2](#), we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.4.2. Let $k \in \mathbb{N}$ be given. A set $E \subseteq \mathbb{R}^k$ defined by

$$E := \left\{ \mathbf{x} = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k \mid a_j \leq x_j \leq b_j \text{ for all } j = 1, 2, \dots, k \right\}$$

is called an **k -cell**, where $-\infty < a_j \leq b_j < +\infty$ for all $j = 1, 2, \dots, k$.

Definition 2.4.3. Let (X, d) be a metric space. For any $p \in X$ and $r \in \mathbb{R}^+$, we define the **neighborhood** of p with radius r as

$$N_r(p) := \{q \in X \mid d(p, q) < r\}.$$

Also, we define the **deleted neighborhood** of p with radius r as

$$\tilde{N}_r(p) := N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}.$$

Especially, if $X = \mathbb{R}^n$ and d is the standard metric on $\mathbb{R}^n$ for some $n \in \mathbb{N}$, then $N_r(p)$ is called the **open ball** of radius r centered at p , which is denoted by $B_r(p)$.

Definition 2.4.4. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if for any $r \in \mathbb{R}^+$, we have $E \cap \tilde{N}_r(p) \neq \emptyset$.
- p is called a **isolated point** of E if there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \cap E = \{p\}$.
- p is called an **interior point** of E if there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \subseteq E$.

Definition 2.4.5. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **open** if any point of E is an interior point of E .
- E is said to be **closed** if any limit point of E belongs to E .

Definition 2.4.6. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **bounded** if there exists some $p \in X$ and $r \in \mathbb{R}^+$ such that $E \subseteq N_r(p)$.
- E is said to be **perfect** if any point of E is a limit point of E .
- E is said to be **dense** if any point of X is either a limit point of E or a point of E .

Theorem 2.4.7. Let (X, d) be a metric space. For any $p \in X$ and $r \in \mathbb{R}^+$, the neighborhood $N_r(p)$ is open.

Proof. Let $q \in N_r(p)$ be given. Then we have $d(p, q) < r \implies \varepsilon := r - d(p, q) > 0$. Let $s \in N_\varepsilon(q)$ be given. Then we have

$$d(q, s) < \varepsilon \implies d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r.$$

Thus $s \in N_r(p)$, which implies that $N_\varepsilon(q) \subseteq N_r(p)$. Since q is arbitrary, $N_r(p)$ is open. $\square$

Theorem 2.4.8. Let (X, d) be a metric space and $E \subseteq X$. Let p be a limit point of E . Then $E \cap N_r(p)$ is infinite for all $r \in \mathbb{R}^+$.

Proof. Assume that there exists some $r \in \mathbb{R}^+$ such that $E \cap N_r(p)$ is finite. Let S be a set defined by

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \tilde{N}_r(p).$$

Since p is a limit point of E , we have $S \neq \emptyset$. Since $E \cap N_r(p)$ is finite, $S \subseteq E \cap N_r(p)$ is also finite by [Theorem 2.2.2](#). Let T be a set defined by

$$T := \{d(p, q) \mid q \in S\} \neq \emptyset.$$

Since $p \notin S$, we have $0 \notin T$. Let $f : S \rightarrow T$ be a function defined by

$$f(q) = d(p, q).$$

Then f is surjective. By [Theorem 2.2.4](#), T is finite. Then, by [Theorem 2.2.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. Then we have

- $N_\varepsilon(p) \subset N_r(p)$.
- $E \cap \tilde{N}_\varepsilon(p) = \emptyset$.

This contradicts the assumption that p is a limit point of E . Therefore, $E \cap N_r(p)$ is infinite. $\square$

Theorem 2.4.9. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \text{ is open} \iff E^c \text{ is closed.}$$

Proof. ($\implies$) Assume that E is open. Let p be a limit point of E^c . Then, for any $r \in \mathbb{R}^+$, we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset.$$

If $p \in E$, then there exists some $\varepsilon \in \mathbb{R}^+$ such that $N_\varepsilon(p) \subseteq E$. Then we have

$$E^c \cap \tilde{N}_\varepsilon(p) = \emptyset,$$

which is a contradiction. Therefore, we have $p \in E^c$, which implies that E^c is closed.

($\impliedby$) Assume that E^c is closed. Let p be a point of E . If p is not an interior point of E , then for any $r \in \mathbb{R}^+$, we have

$$E^c \cap N_r(p) \neq \emptyset.$$

Since $p \in E \implies p \notin E^c$, we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that p is a limit point of E^c . Since E^c is closed, we have $p \in E^c$, which is a contradiction. Therefore, p is an interior point of E . Since p is arbitrary, E is open. $\square$

Corollary 2.4.10. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \text{ is closed} \iff E^c \text{ is open.}$$

Theorem 2.4.11. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be an open set for each $y \in Y$. Then we have*

$$\bigcup_{y \in Y} E_y \text{ is open}$$

whether Y is finite, countable, or uncountable. This theorem states that an arbitrary union of open sets is open.

Proof. Let $p \in \bigcup_{y \in Y} E_y$ be given. Then there exists some $y_0 \in Y$ such that $p \in E_{y_0}$.

Since E_{y_0} is open, there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \subseteq E_{y_0}$. Then we have

$$N_r(p) \subseteq E_{y_0} \subseteq \bigcup_{y \in Y} E_y,$$

which implies that p is an interior point of $\bigcup_{y \in Y} E_y$. Since p is arbitrary, $\bigcup_{y \in Y} E_y$ is open. $\square$

Corollary 2.4.12. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be a closed set for each $y \in Y$. Then we have*

$$\bigcap_{y \in Y} E_y \text{ is closed}$$

whether Y is finite, countable, or uncountable. This corollary states that an arbitrary intersection of closed sets is closed.

Proof. By Corollary 2.4.10, $(E_y)^c$ is open for each $y \in Y$. Then, by Theorem 2.4.11, we have

$$\bigcup_{y \in Y} (E_y)^c \text{ is open.}$$

By Theorem 2.4.9, we have

$$\bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed.}$$

$\square$

Theorem 2.4.13. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be open sets for some $n \in \mathbb{N}$. Then we have*

$$\bigcap_{k=1}^n E_k \text{ is open.}$$

This theorem states that a finite intersection of open sets is open.

Proof. Let $p \in \bigcap_{k=1}^n E_k$ be given. Then we have $p \in E_j$ for all $j = 1, 2, \dots, n$.

Since E_j is open, there exists some $r_j \in \mathbb{R}^+$ such that $N_{r_j}(p) \subseteq E_j$ for each $j = 1, 2, \dots, n$. Let R be a set defined by

$$R := \{r_j \mid j = 1, 2, \dots, n\} \subseteq \mathbb{R}^+.$$

Let $f : J_n \rightarrow R$ be a function defined by

$$f(j) = r_j.$$

Then f is surjective. By [Lemma 2.2.1](#) and [theorem 2.2.4](#), R is finite. Then, by [Theorem 2.2.5](#), R has a minimum element, say $r = \min R > 0$. Then we have

$$N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \text{ for all } j = 1, 2, \dots, n \implies N_r(p) \subseteq \bigcap_{k=1}^n E_k,$$

which implies that p is an interior point of $\bigcap_{k=1}^n E_k$. Since p is arbitrary, $\bigcap_{k=1}^n E_k$ is open. $\square$

Corollary 2.4.14. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be closed sets for some $n \in \mathbb{N}$. Then we have*

$$\bigcup_{k=1}^n E_k \text{ is closed.}$$

This corollary states that a finite union of closed sets is closed.

Proof. By [Corollary 2.4.10](#), $(E_k)^c$ is open for each $k = 1, 2, \dots, n$. Then, by [Theorem 2.4.13](#), we have

$$\bigcap_{k=1}^n (E_k)^c \text{ is open.}$$

By [Theorem 2.4.9](#), we have

$$\bigcup_{k=1}^n E_k = \left(\bigcap_{k=1}^n (E_k)^c \right)^c \text{ is closed.}$$

$\square$

Definition 2.4.15. Let (X, d) be a metric space and $E \subseteq X$. Let E' be the set of all limit points of E . The **closure** of E , which is denoted by $\overline{E}$, is defined by

$$\overline{E} := E \cup E'.$$

Theorem 2.4.16. *Let (X, d) be a metric space and $E \subseteq X$. Then we have the following properties:*

(A) $\overline{E}$ is closed.

(B) $E = \overline{E} \iff E$ is closed.

(C) For any closed subset $F \subseteq X$ such that $E \subseteq F$, we have $\overline{E} \subseteq F$.

(C) states that the closure of E is the smallest closed set containing E .

Proof. (A) We will show that $(\overline{E})^c$ is open. Let $p \in (\overline{E})^c$ be given. Since $p \notin E$ and $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies E \cap N_r(p) = \emptyset \implies N_r(p) \subseteq E^c.$$

By Theorem 2.4.7, $N_r(p)$ is open. Then, for any $q \in N_r(p)$, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$\begin{aligned} N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c &\implies E \cap N_\varepsilon(q) = \emptyset \\ &\implies E \cap (\tilde{N}_\varepsilon(q) \cup \{q\}) = \emptyset \\ &\implies (E \cap \tilde{N}_\varepsilon(q)) \cup (E \cap \{q\}) = \emptyset \\ &\implies E \cap \tilde{N}_\varepsilon(q) = \emptyset \\ &\implies q \notin E'. \end{aligned}$$

Since q is arbitrary, we have

$$N_r(p) \subseteq (E')^c \implies N_r(p) \subseteq (\overline{E})^c = E^c \cap (E')^c,$$

which implies that p is an interior point of $(\overline{E})^c$. Since p is arbitrary, $(\overline{E})^c$ is open, which implies that $\overline{E}$ is closed by Corollary 2.4.10.

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$.

(C) Let $F \subseteq X$ be a closed set such that $E \subseteq F$. Let $p \in E'$ be given. Then, for any $r \in \mathbb{R}^+$, we have

$$E \cap \tilde{N}_r(p) \neq \emptyset \implies F \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that $p \in F' \subseteq F$. Therefore, we have $E' \subseteq F$. Since $E \subseteq F$, we have $\overline{E} = E \cup E' \subseteq F$. □

Theorem 2.4.17. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. Then we have

$$\sup E \in \overline{E},$$

which implies that

$$E \text{ is closed} \implies \sup E \in E.$$

Proof. By the *least upper bound property*, there exists $\alpha = \sup E \in \mathbb{R}$. Assume that $\alpha \in E$. Then we have

$$E \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

Otherwise, for any $x \in \mathbb{R}^+$, $\alpha - x$ is not an upper bound of E . Thus there exists some $p \in E$ such that

$$\alpha - x < p < \alpha \implies p \in \tilde{N}_x(\alpha) \cap E,$$

which implies that $\alpha \in E'$. Therefore, we have

$$E' \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

□

Theorem 2.4.18. Let (X, d) be a metric space and $E \subseteq X$. Then we have

$$E \setminus E' = \{p \in X \mid p \text{ is an isolated point of } E\}.$$

Proof. ($\subseteq$) Let $p \in E \setminus E'$ be given. Since $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies (E \cap N_r(p)) \setminus \{p\} = \emptyset.$$

Since $p \in E$ and $p \in N_r(p)$, we have

$$E \cap N_r(p) = \{p\},$$

which implies that p is an isolated point of E .

($\supseteq$) Let p be an isolated point of E . Then there exists some $r \in \mathbb{R}^+$ such that

$$E \cap N_r(p) = \{p\}.$$

Then we have

- $E \cap \tilde{N}_r(p) = \emptyset \implies p \notin E'$.
- $p \in E$.

Therefore, we have $p \in E \setminus E'$.

□

Definition 2.4.19. Let (X, d) be a metric space and $E \subseteq X$. The **boundary** of E , which is denoted by ∂E , is defined by

$$\partial E := \overline{E} \cap \overline{E^c}.$$

Lemma 2.4.20. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$. Then we have*

$$\tilde{N}_r(\mathbf{p}) \neq \emptyset \quad \text{for all } \mathbf{p} \in \mathbb{R}^n \text{ and } r \in \mathbb{R}^+.$$

Proof. Let $\mathbf{p} = (p_1, p_2, \dots, p_n) \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ be given. Then we have

$$d\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1, \mathbf{p}\right) = \left\|\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) - \mathbf{p}\right\| = \left\|\frac{r}{2}\mathbf{e}_1\right\| = \frac{r}{2}\|\mathbf{e}_1\| = \frac{r}{2} < r.$$

where $\mathbf{e}_1 = (1, 0, 0, \dots, 0) \in \mathbb{R}^n$. Therefore, we have

$$\mathbf{p} \neq \left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) \in \tilde{N}_r(\mathbf{p}) \implies \tilde{N}_r(\mathbf{p}) \neq \emptyset.$$

□

Theorem 2.4.21. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)'),$$

which states that the boundary of E consists the following three disjoint sets:

- *The set of all isolated points of E .*
- *The set of all isolated points of E^c .*
- *The set of all limit points of both E and E^c .*

Proof. By the definition of the boundary, we have

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= (E \cup E') \cap (E^c \cup (E^c)') \\ &= (E \cap E^c) \cup (E \cap (E^c)') \cup (E' \cap E^c) \cup (E' \cap (E^c)'). \end{aligned}$$

Consider the following equivalences:

- $E = (E \setminus E') \cup (E \cap E')$.
- $E^c = (E^c \setminus (E^c)') \cup (E^c \cap (E^c)').$

Then we have the following equalities:

- $E \cap (E^c)' = ((E \setminus E') \cap (E^c)') \cup ((E \cap E') \cap (E^c)').$
- $E' \cap E^c = (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c \cap (E^c)')).$

Consider the last two terms of the above equalities:

- $((E \cap E') \cap (E^c)') \subseteq E' \cap (E^c)'.$
- $(E' \cap (E^c \cap (E^c)')) \subseteq E' \cap (E^c)'.$

Then we have

$$\begin{aligned}\partial E &= \emptyset \cup ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \\ &= ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)')\end{aligned}$$

For any $\mathbf{p} \in E \setminus E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(\mathbf{p}) = \emptyset \implies \tilde{N}_r(\mathbf{p}) \subseteq E^c.$$

By Lemma 2.4.20, we have

$$\tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) = \tilde{N}_{\min(s,r)}(\mathbf{p}) \neq \emptyset \quad \text{for all } s \in \mathbb{R}^+.$$

Then, for any $s \in \mathbb{R}^+$, we have

$$\begin{aligned}\tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) \neq \emptyset &\implies \tilde{N}_s(\mathbf{p}) \cap (\tilde{N}_r(\mathbf{p}) \cap E^c) \neq \emptyset \\ &\implies \emptyset \neq (\tilde{N}_s(\mathbf{p}) \cap E^c) \cap \tilde{N}_r(\mathbf{p}) \subseteq \tilde{N}_s(\mathbf{p}) \cap E^c \\ &\implies E^c \cap \tilde{N}_s(\mathbf{p}) \neq \emptyset,\end{aligned}$$

which implies that $\mathbf{p} \in (E^c)'$. Then we have

$$E \setminus E' \subseteq (E^c)' \implies (E \setminus E') \cap (E^c)' = E \setminus E'.$$

Similarly, we can show that

$$E^c \setminus (E^c)' \subseteq E' \implies E' \cap (E^c \setminus (E^c)') = E^c \setminus (E^c)'.$$

Therefore, we have

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)').$$

□

Remark. Let (X, d) be a metric space and $Y \subseteq X$. Then, $(Y, d|_{Y \times Y})$ is also a metric space, since

$$d|_{Y \times Y} : Y \times Y \rightarrow \mathbb{R}$$

satisfies the properties of a metric. We call $(Y, d|_{Y \times Y})$ a **subspace** of (X, d) . For the sake of simplicity, we will denote $d|_{Y \times Y}$ by d and write (Y, d) instead of $(Y, d|_{Y \times Y})$.

Definition 2.4.22. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in the subspace (Y, d) . That is, for any $p \in E$, there exists some $r \in \mathbb{R}^+$ such that

$$Y \cap N_r(p) \subseteq E.$$

Theorem 2.4.23. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is open relative to Y if and only if there exists an open set V in X such that $E = V \cap Y$.*

Proof. ($\implies$) Let E be an open set relative to Y . Then, for each $p \in E$, there exists some $r_p \in \mathbb{R}^+$ such that

$$Y \cap N_{r_p}(p) \subseteq E.$$

Let V be a set defined by

$$V := \bigcup_{p \in E} N_{r_p}(p).$$

By Theorems 2.4.7 and 2.4.11, V is open in X . Since $p \in N_{r_p}(p) \subseteq V$ for each $p \in E$, we have

- $E \subseteq Y$ and $E \subseteq V \implies E \subseteq V \cap Y$.
- $\bigcup_{p \in E} (Y \cap N_{r_p}(p)) \subseteq E \implies Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \implies Y \cap V \subseteq E$.

Therefore, we have $E = V \cap Y$.

($\impliedby$) Let V be an open set in X such that $E = V \cap Y$. Then, for each $p \in E \subseteq V$, there exists some $r_p \in \mathbb{R}^+$ such that

$$N_{r_p}(p) \subseteq V \implies Y \cap N_{r_p}(p) \subseteq V \cap Y = E.$$

□

2.5 Compact Sets

Let (X, d) be a metric space and $E \subseteq X$. Let A be a set and let G_α be an open set in X for each $\alpha \in A$. Then the collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of E if

- $S \subseteq A$.
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$.

E is called **compact** if any open cover of E has a finite subcover. That is, for each open cover $\{G_\alpha\}_{\alpha \in A}$ of E , there exists a subset $S_A \subseteq A$ such that

- S_A is finite.
- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$.

Theorem 2.5.1. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is compact relative to Y if and only if E is compact relative to X .*

Proof. ($\implies$) Let E be a compact set relative to Y . Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of E relative to X . Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. By Theorem 2.4.23, H_α is open relative to Y . Since $E \subseteq Y$, we have

$$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in A}$ is an open cover of E relative to Y . Since E is compact relative to Y , there exists a finite subcover $\{H_\alpha\}_{\alpha \in S_A}$ of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to X . Therefore, E is compact relative to X .

($\impliedby$) Let E be a compact set relative to X . Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By Theorem 2.4.23, there exists an open set G_α in X such that $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E relative to X . Since E is compact relative to X , there exists a finite subcover $\{G_\alpha\}_{\alpha \in S_A}$ of E relative to X . Then we have

$$E \subseteq \left(\bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to Y . Therefore, E is compact relative to Y . $\square$

Remark. The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

“Let (X, d) be a metric space and $E \subseteq X$.”

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- E is open. $\implies E$ is open relative to X .
- E is closed. $\implies E$ is closed relative to X .
- E is compact. $\implies E$ is compact relative to X .

if there is no specific mention of the subspace topology.

Theorem 2.5.2. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is closed.*

Proof. Let E be compact. We will prove that E^c is open, which implies that E is closed by [Corollary 2.4.10](#). Let $p \in E^c$ be given. Since $p \notin E$, we have

$$r_q := \frac{d(p, q)}{2} > 0 \quad \text{for each } q \in E.$$

Since $q \in N_{r_q}(q)$ for each $q \in E$, we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \{N_{r_q}(q)\}_{q \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{N_{r_q}(q)\}_{q \in S}$ of E . Let T be a set defined by

$$T := \{r_q \mid q \in S\}.$$

Let $f : S \rightarrow T$ be a function defined by

$$f(q) = r_q.$$

Then f is surjective. By [Theorem 2.2.4](#), T is finite. Then, by [Theorem 2.2.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. For any $x \in N_\varepsilon(p)$, we have

$$d(x, p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some $q \in S$ such that $x \in N_{r_q}(q)$. Then we have

$$d(p, q) \leq d(p, x) + d(x, q) < r_q + r_q = 2r_q = d(p, q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \emptyset$ for each $q \in S$.
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$.

Therefore, we have $N_\varepsilon(p) \subseteq E^c$. Since p is arbitrary, E^c is open. $\square$

Theorem 2.5.3. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is bounded.*

Proof. Let E be compact. Let $p \in E$ be given. Let $\{G_n\}_{n \in \mathbb{N}}$ be a collection of open sets defined by

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \leq m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For any $q \in X$, by [Theorem 1.5.1](#), there exists $n \in \mathbb{N}$ such that $n > d(p, q)$. Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n \in \mathbb{N}} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_n\}_{n \in S}$ of E . Since $S \subseteq \mathbb{N}$ is finite, by [Theorem 2.2.5](#), S has a greatest element, say $m = \max S$. Then we have

$$E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p),$$

which implies that E is bounded. $\square$

Theorem 2.5.4. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then any closed subset $F \subseteq E$ is compact.*

Proof. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of F . We have the following statements:

- F is closed. $\implies F^c$ is open.
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$.

Then we have

$$X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

$$\bullet E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

By Theorem 2.4.11, $H_\alpha \cup F^c$ is open for each $\alpha \in A$. Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{H_\alpha \cup F^c\}_{\alpha \in B}$ of E . Then we have

$$\begin{aligned} \bullet E &\subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c). \\ \bullet F &\subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c. \end{aligned}$$

Since $F \cap F^c = \emptyset$, we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in B}$ is a finite subcover of F . Therefore, F is compact. $\square$

Theorem 2.5.5. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then any infinite subset of E has a limit point in E .*

Proof. Let E be compact. Let $F \subseteq E$ be an infinite subset of E . Assume that F has no limit point in E . Then, for each $p \in E$, there exists $r_p \in \mathbb{R}^+$ such that

$$\tilde{N}_{r_p}(p) \cap F = \emptyset \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let $\{G_\alpha\}_{\alpha \in E}$ be a collection of open sets defined by

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since $\alpha \in G_\alpha$ for each $\alpha \in E$, we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S}$ of E . Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since $F \subseteq \bigcup_{\alpha \in S} G_\alpha$, we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by Theorem 2.2.2. Therefore, any infinite subset of E has a limit point in E . $\square$

Theorem 2.5.6. *Let (X, d) be a metric space and let $\{K_\alpha\}_{\alpha \in A}$ be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset \text{ for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Proof. Assume that $\bigcap_{\alpha \in A} K_\alpha = \emptyset$. Then we have

$$X = \left(\bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each K_α is compact, K_α is closed by [Theorem 2.5.2](#). Then, by [Corollary 2.4.10](#), we have

$$\text{Each } (K_\alpha)^c \text{ is open} \implies \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X.$$

Since $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$, we have

$$\{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha.$$

Since K_α is compact, there exists a finite subcover $\{(K_\beta)^c\}_{\beta \in B_\alpha}$ of K_α . Then we have

$$\begin{aligned} K_\alpha &\subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta \right)^c \\ &\implies K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta \right) = \emptyset \\ &\implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \emptyset. \end{aligned}$$

Since $\{\alpha\}$ and B_α are finite, $B_\alpha \cup \{\alpha\}$ is also finite by [Theorem 2.2.11](#), which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

□

Lemma 2.5.7. *Let $(a_n)_{n=1}^\infty$ and $(b_n)_{n=1}^\infty$ be sequences in $\mathbb{R}$, and let $\{I_n\}_{n=1}^\infty$ be a sequence of intervals in $\mathbb{R}$ such that*

- $I_n = [a_n, b_n]$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$

where $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Let E be a set defined by

$$E := \{a_n \mid n \in \mathbb{N}\}.$$

Since E is bounded above by b_1 , there exists $\alpha = \sup E$. Then we have

$$a_m \leq \alpha \quad \text{for all } m \in \mathbb{N}.$$

For any $m \in \mathbb{N}$, we have b_m is an upper bound of E since

- $a_n \leq a_m \leq b_m$ for all $n \in \mathbb{N}$ such that $n \leq m$.
- $a_n \leq b_n \leq b_m$ for all $n \in \mathbb{N}$ such that $n > m$.

Therefore, we have

$$a_m \leq \alpha \leq b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

Theorem 2.5.8. Let $k \in \mathbb{N}$ be given. Let $\{a_{n,j}\}_{n=1}^{\infty}$ and $\{b_{n,j}\}_{n=1}^{\infty}$ be sequences in $\mathbb{R}$ for all $j = 1, 2, \dots, k$, and let $\{I_n\}_{n=1}^{\infty}$ be a sequence of k -cells in $\mathbb{R}^k$ such that

- $I_n = \{\mathbf{x} \in \mathbb{R}^k \mid a_{n,j} \leq x_j \leq b_{n,j} \text{ for all } j = 1, 2, \dots, k\}$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$

where $a_{n,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Let $\{I_{n,j}\}_{n=1}^{\infty}$ be a sequence of intervals in $\mathbb{R}$ defined by

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Then we have $I_{n+1,j} \subseteq I_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. By [Lemma 2.5.7](#), we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \emptyset \quad \text{for all } j = 1, 2, \dots, k.$$

Let $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$ be given for all $j = 1, 2, \dots, k$. Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k) \in \mathbb{R}^k$. Then we have

$$\alpha \in I_n \quad \text{for all } n \in \mathbb{N} \implies \alpha \in \bigcap_{n=1}^{\infty} I_n.$$

□

Theorem 2.5.9. Let $(\mathbb{R}^k, \|\cdot\|)$ be given for some $k \in \mathbb{N}$. Then any k -cell is compact.

Proof. Let I be a k -cell in $\mathbb{R}^k$, and let $I_1 := I$. Then we have

$$I_1 = \prod_{j=1}^k [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where $-\infty < a_{1,j} \leq b_{1,j} < +\infty$ for all $j = 1, 2, \dots, k$. Let $I_{1,j,m}$ be an interval in $\mathbb{R}$ defined by

$$I_{1,j,m} := \begin{cases} \left[a_{1,j}, \frac{a_{1,j} + b_{1,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{1,j} + b_{1,j}}{2}, b_{1,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Let S_k be a set defined by

$$S_k := \{1, 2\}^k = \left\{ \mathbf{m} \in \mathbb{R}^k \mid m_j \in \{1, 2\} \text{ for all } j = 1, 2, \dots, k \right\}.$$

By [Theorem 2.2.13](#), S_k is finite. We have

$$I_1 = \bigcup_{\mathbf{m} \in S_k} \left(\prod_{j=1}^k I_{1,j,m_j} \right).$$

Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of I_1 . Assume that I_1 has no finite subcover. Then, for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{1,j,m_j}$. Assume that for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha \in A_{\mathbf{m}}}.$$

Then we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \quad \text{for all } \mathbf{m} \in S_k,$$

where $\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha$ is open by [Theorem 2.4.11](#). Then we have

$$I_1 \subseteq \bigcup_{\mathbf{m} \in S_k} \left(\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha,$$

where $A := \bigcup_{\mathbf{m} \in S_k} A_{\mathbf{m}}$ is finite by [Theorem 2.2.11](#). Thus I_1 has a finite subcover, which is a contradiction. Therefore, there exists $\mathbf{m}_1 \in S_k$ such that

$$\prod_{j=1}^k I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let I_2 be a k -cell in $\mathbb{R}^k$ defined by

$$I_2 := \prod_{j=1}^k [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined by

$$\begin{aligned} \bullet \quad a_{2,j} &:= \begin{cases} a_{1,j} & \text{if } (m_1)_j = 1 \\ \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 2 \end{cases} \\ \bullet \quad b_{2,j} &:= \begin{cases} \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 1 \\ b_{1,j} & \text{if } (m_1)_j = 2 \end{cases} \end{aligned}$$

for all $j = 1, 2, \dots, k$. Then, for each $j = 1, 2, \dots, k$, we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^k I_{1,j,(m_1)_j} \subseteq I_1.$$

Let $I_{2,j,m}$ be an interval in $\mathbb{R}$ defined by

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \frac{a_{2,j} + b_{2,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Then we have

$$I_2 = \bigcup_{\mathbf{m} \in S_k} \left(\prod_{j=1}^k I_{2,j,m_j} \right).$$

We have shown that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of I_2 , and I_2 has no finite subcover. Then, for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{2,j,m_j}$. Therefore, there exists $\mathbf{m}_2 \in S_k$ such that

$$\prod_{j=1}^k I_{2,j,(\mathbf{m}_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence $(\mathbf{m}_n)_{n=1}^\infty$ in S_k .
- The sequences $(a_{n,j})_{n=1}^\infty$ and $(b_{n,j})_{n=1}^\infty$ in $\mathbb{R}$ for all $j = 1, 2, \dots, k$.
- The sequence $(I_{n,j,m})_{n=1}^\infty$ of intervals in $\mathbb{R}$ for all $j = 1, 2, \dots, k$ and $m = 1, 2$.
- The sequence $(I_n)_{n=1}^\infty$ of k -cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \frac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^k (b_{1,j} - a_{1,j})^2}$. Then we have

$$\|x - y\| \leq \sqrt{\sum_{j=1}^k (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } x, y \in I_n \text{ and } n \in \mathbb{N}.$$

By [Theorem 2.5.8](#), we have $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. Let $\beta \in \bigcap_{n=1}^{\infty} I_n$ be given. Then there exists $\alpha_0 \in A$ such that $\beta \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\beta) \subseteq G_{\alpha_0}.$$

By [Corollary 1.5.2](#), there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\beta \in I_N$, we have

$$\|\beta - \gamma\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \gamma \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\beta) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. $\square$

Corollary 2.5.10. *For any $-\infty < a \leq b < +\infty$, the closed interval $[a, b]$ is compact in $(\mathbb{R}, \|\cdot\|)$.*

Theorem 2.5.11. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$E \text{ is compact} \iff E \text{ is closed and bounded.}$$

*This theorem is called the **Heine-Borel Theorem**.*

Proof. ($\implies$) Let E be compact. E is closed by [Theorem 2.5.2](#), and E is bounded by [Theorem 2.5.3](#).

($\impliedby$) Let E be closed and bounded. Then there exists some $p \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(p)$. Let I be a n -cell in $\mathbb{R}^n$ defined by

$$I := \prod_{j=1}^n [p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(p) \subseteq I$. By [Theorem 2.5.9](#), I is compact. Since E is closed, by [Theorem 2.5.4](#), E is compact. $\square$

2.6 Connected Sets

Let (X, d) be a metric space and $E \subseteq X$. E is called **connected** if there exists no pair of sets $A, B \subseteq X$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.1. *Let $(\mathbb{R}, \|\cdot\|)$ be given. Then $E \subseteq \mathbb{R}$ is connected if and only if*

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Let $E \subseteq \mathbb{R}$ be connected. Assume that there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Let A, B be sets defined by

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \emptyset$ and $B \neq \emptyset$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\impliedby$) Let $E \subseteq \mathbb{R}$ be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.

- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$ be given. Without loss of generality, assume that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \emptyset$.
- $A \cap [a, b]$ is bounded above by b .

Then the *existence* of α is guaranteed by the *least upper bound property*. By [Theorem 2.4.17](#), we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By [Definition 2.4.4](#), it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\tilde{N}_r(\alpha) \cap B = \emptyset \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Let (X, d) be a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \rightarrow X$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

For any $\varepsilon \in \mathbb{R}^+$, there exists $N \in \mathbb{N}$ such that $n \geq N \implies d(x_n, x) < \varepsilon$.

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x .
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim_{n \rightarrow \infty} x_n = x$.
- $x_n \rightarrow x$ as $n \rightarrow \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.1. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $(x_n)_{n=1}^{\infty}$ be a sequence defined by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

Let $(\mathbb{R}^+, \|\cdot\|)$ be a subspace of $(\mathbb{R}, \|\cdot\|)$. Let $(x_n)_{n=1}^\infty$ be a sequence defined by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By [Theorem 1.5.1](#), for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \frac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left| \frac{1}{n} - y \right| > \left| \frac{1}{N_y} - y \right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^\infty$ diverges in $(\mathbb{R}^+, \|\cdot\|)$.

Remark. Let (X, d) be a metric space. In this textbook, $(x_n)_{n=1}^\infty$ represents a sequence in (X, d) , while $\{x_n\}_{n=1}^\infty$ represents a set defined by

$$\{x_n\}_{n=1}^\infty := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.2. Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^\infty$ is said to be **Bounded** if $\{x_n\}_{n=1}^\infty$ is bounded.

Theorem 3.1.3. Let (X, d) be a metric space. Let $(x_n)_{n=1}^\infty$ be a convergent sequence. Then we have

(A) $(x_n)_{n=1}^\infty$ is bounded.

(B) The limit of $(x_n)_{n=1}^\infty$ is unique.

Proof. Let x be the limit of $(x_n)_{n=1}^\infty$.

(A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Let S be a set defined by

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : J_N \rightarrow S$ be a function defined by

$$f(k) = d(x_k, x) \quad \text{for all } k \in J_N.$$

Then f is surjective. By [Lemma 2.2.3](#), S is finite. Then, by [Theorem 2.2.5](#), S has a greatest element, say $M = \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^\infty$ is bounded.

(B) Let y be the limit of $(x_n)_{n=1}^{\infty}$. Assume that $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \frac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \frac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \frac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

□

Theorem 3.1.4. Let (X, d) be a metric space. Then the following statements hold:

(A) A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if

For any $\varepsilon \in \mathbb{R}^+$, $N_{\varepsilon}(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Let $E \subseteq X$ be given. If x is a limit point of E , then there exists a sequence $(x_n)_{n=1}^{\infty}$ in (X, d) such that

$$\lim_{n \rightarrow \infty} x_n = x.$$

Proof. (A) ($\implies$) Let $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_{\varepsilon} \in \mathbb{N}$ such that

$$n \geq M_{\varepsilon} \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_{\varepsilon}}^{\infty} \subseteq N_{\varepsilon}(x).$$

($\impliedby$) Let $(x_n)_{n=1}^{\infty}$ be a sequence in (X, d) such that

For any $\varepsilon \in \mathbb{R}^+$, $N_{\varepsilon}(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Let $E \subseteq X$ be given. Assume that x is a limit point of E . Then for any $\varepsilon \in \mathbb{R}^+$, we have $(N_{\varepsilon}(x) - \{x\}) \cap E \neq \emptyset$. Thus we can choose a sequence $(y_n)_{n=1}^{\infty}$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n \rightarrow \infty} y_n = x$.

□

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- 3.5 Some Special Sequences**
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Appendix: Construction of $\mathbb{R}$

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